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LiDAR-Simulated Multimodal and Self-Supervised Contrastive Digital Twin Approach for Probabilistic Point Cloud Generation of Rail Fasteners

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Abstract: This study presents a novel deep-learning framework designed to efficiently generate high-fidelity three-dimensional (3D) point clouds of rail fasteners. The proposed method overcomes limitations associated with traditional light detection and ranging (LiDAR) technology, including high cost and incomplete data acquisition. The framework utilizes transformers and Bayesian neural networks for self-supervised contrastive learning, enabling real-time 3D point cloud generation from depth and grayscale images. This approach fosters automation and efficiency in maintenance tasks by producing accurate geometric reconstructions of rail fasteners. Beyond 3D generation, the framework learns latent representations of two-dimensional (2D) data, potentially impacting downstream tasks. This research contributes significantly to the development of digital twins in infrastructure monitoring by providing a cost-effective and efficient pathway for real-time 3D data generation. The proposed methodology holds promise for enhancing safety, reliability, and cost-effectiveness across a broad spectrum of infrastructure applications. **DOI: 10.1061/JCCEE5.CPENG-6137.** *© 2025 American Society of Civil Engineers.*

Author keywords: Simulate light detection and ranging (LiDAR); Three-dimensional (3D) point cloud modeling; Self-supervised contrastive learning; Bayesian neural networks; Digital twin; Infrastructure monitoring.

Introduction

In recent years, the fields of deep learning and 3D reconstruction have seen remarkable advancements, leading to significant breakthroughs and practical applications across various industries. One sector that has notably reaped the benefits of these innovations is civil infrastructure inspection, where regular monitoring of railway infrastructure is paramount for ensuring construction quality, safety, and maintenance (Spencer et al. 2019). Central to this assurance is the regular inspection of critical components, among which rail fasteners play a pivotal role (see Fig. 1). These fasteners secure rails to the base and are crucial for maintaining track gauges. Any failure in these components can lead to catastrophic derailments and accidents (Akbari and Cammi 2024). In this regard, light detection and ranging (LiDAR) technology has played a crucial role in this domain by providing high-accuracy 3D point clouds for inspection (Prieto et al. 2024). Despite its effectiveness, the high cost and complexity of LiDAR systems have limited their widespread use, particularly in smaller railway networks. Consequently, there has been growing interest in exploring alternative solutions, such as image-based approaches and deep-learning techniques, to overcome these limitations (Chastre et al. 2023).

The journey of 3D reconstruction technology began with foundational developments in computer vision and graphics, aimed at modeling and visualizing three-dimensional objects and environments. This technology extends beyond mere visual representation, offering precise measurements, positional data, and range information (Kim et al. 2022). Such detailed information is invaluable not only for inspection and analysis but also for creating "digital twins"—highly accurate 3D models integrated with supplementary data (Hoskere et al. 2022). These digital twins enable human interaction and even predictive analysis through simulations via 3D replicas of the main component (Agarwal et al. 2021). Historically, techniques such as structure from motion (SfM) (Nassar and Jung 2012) and multiview stereo (MVS) (Ma et al. 2023) have been instrumental in 3D reconstruction. These methods have continually improved in terms of accuracy and robustness, supported by a range of open-source and commercial software tools (Hosken 2021; Kattan et al. 2022) that provide comprehensive solutions for image-based 3D reconstruction (Maalek and Lichti 2021; Nyimbili et al. 2016; Pepe and Costantino 2021; Sani et al. 2022; Xie et al. 2023).

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Fig. 1. High-speed railways fastener.

The integration of convolutional neural networks (CNNs) post-2015 marked a significant shift, with MVSNet exemplifying the impact of deep learning on 3D reconstruction (Yao et al. 2018). The integration of object detection and semantic segmentation within reconstruction processes promises further improvements in accuracy and efficiency (Han et al. 2021). Challenges remain, including achieving high accuracy in complex scenes and scaling for large data sets. Real-time reconstruction for dynamic scenes, crucial for augmented reality and robotics, is a key research focus. Integration with sensor fusion (LiDAR, radar) and semantic understanding is expected to expand 3D reconstruction's applications across various fields.

Deep learning has revolutionized image-based 3D reconstruction, overcoming the limitations of traditional methods (Lu et al. 2024). Early CNN-based techniques such as temporally invariant learned detector (TILDE) (Verdie et al. 2015) and DeepDesc (Simo-Serra et al. 2015) initially focused on feature extraction and matching but were limited in accuracy and efficiency. To enhance point cloud quality, supervised learning approaches such as PointCleanNet (Rakotosaona et al. 2020) emerged, effectively denoising point clouds—critical for accurate reconstructions. However, these methods rely on large labeled data sets, which are costly and time-consuming. Unsupervised alternatives such as total denoising (Casajus et al. 2019) offer promise but struggle to match supervised performance. Advances in deep learning have also improved semantic and instance segmentation through techniques such as PointNet (Chen et al. 2019) and Pix2Vox (Xie et al. 2019), enhancing the understanding of object shapes and relationships within scenes for more precise 3D reconstructions. Yet, challenges remain in generalizing these techniques to diverse object shapes and materials, especially in complex environments (Xu et al. 2021). The extraction of 2D information from images has further strengthened reconstruction methods, with techniques such as 3D-R2N2 (Leibe et al. 2016; Hong et al. 2022) utilizing neural networks to generate 3D representations from 2D data. However, reconstructing occluded or challenging viewpoints remains difficult, prompting the need for better integration of geometric priors and multiview data. Implicit neural representations such as NeRF (Remondino et al. 2023) show promise for efficient, continuous geometry modeling, prioritizing speed, quality, and scalability, though they require substantial computational resources. Data fusion, integrating information from multiple sensors such as cameras and LiDAR, offers another avenue for enhancing 3D reconstruction accuracy and robustness (Ma and Liu 2018; Li et al. 2023). While this approach compensates for individual sensor limitations and enriches the data, it also introduces complexity, requiring powerful hardware and advanced algorithms to resolve conflicts between sensor data (Cabral et al. 2023).

Convolutional neural networks (CNNs) have automated 3D reconstruction processes, significantly enhancing accuracy and efficiency in infrastructure inspection. Image-based 3D models enable engineers to inspect bridges, buildings, and other structures with greater precision, detecting damage that might elude visual inspection (Zhang 2020). Additionally, 3D reconstruction facilitates construction monitoring, allowing for progress tracking, deviation detection, and optimization of schedules (Son and Kim 2010). In the aftermath of natural disasters, these models are crucial for rapid damage assessment, enabling swift response and recovery (Zhou et al. 2016). The integration of 3D models with sensor data into digital twins offers valuable insights into infrastructure health, supporting predictive maintenance and asset management (Torok et al. 2014). However, challenges such as real-time generation speed, high costs, sensor noise, and complex parameter requirements persist. To address these, continued research into data fusion and deep learning is essential.

Therefore, this paper presents a novel deep-learning methodology for generating high-fidelity 3D point clouds of rail fasteners. This framework transcends the limitations of traditional LiDAR by effectively utilizing ubiquitous data sources—depth and grayscale images. This shift toward readily available data offers a significant cost advantage over existing methods. The core of our contribution lies in the synergistic application of transformers (Islam et al. 2024; Martin-Gutierrez et al. 2021) and Bayesian neural networks (Lampinen and Vehtari 2001; Mullachery et al. 2018; Xue et al. 2022). This powerful combination facilitates self-supervised contrastive learning (Jaiswal et al. 2021; Liu et al. 2023), enabling the network to learn robust representations directly from unlabeled data, a prevalent challenge in civil infrastructure inspection. This innovative approach empowers the framework to remarkably transform low-dimensional depth and grayscale images into real-time, high-resolution 3D point clouds of rail fasteners. Beyond immediate benefits in inspection automation and efficiency, this research offers broader ramifications for the civil engineering computing community. The ability to generate accurate geometric reconstructions paves the way for significant advancements in digital twin technology, a cornerstone of modern infrastructure monitoring. Our framework provides a cost-effective and efficient pathway for generating real-time 3D data, a critical component for building and maintaining accurate digital twins of infrastructure assets. This, in turn, has the potential to revolutionize infrastructure monitoring by enabling proactive maintenance strategies. By facilitating the development of digital twins, the proposed methodology holds immense potential to enhance safety, reliability, and cost-effectiveness across a broad spectrum of civil engineering applications.

Beyond these core contributions, our approach offers several additional advantages:

- *Robustness to environmental variability:* Perform effectively under diverse lighting and weather conditions, making it suitable for real-world deployment. Contrary to Lidar, this generated point cloud ignores the noise and focuses more on the main features.
- *Contributions to downstream tasks:* The learned representations of depth and grayscale depth image information can be utilized for various downstream tasks, such as anomaly detection and predictive maintenance. This versatility expands the utility of our framework beyond 3D point cloud generation, simulating

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the learning capabilities of advanced LiDAR systems for proactive maintenance and enhanced infrastructure management.

The paper unfolds with an introduction, followed by a methodology. The methodology details the novel multimodal architecture's development. Following, the results section showcases the architecture's efficacy in point cloud generation. The discussion contextualizes findings in civil engineering. Finally, the conclusion synthesizes key contributions.

Methodology

This paper proposes a novel deep-learning framework for generating high-fidelity 3D point clouds of rail track fasteners, leveraging depth images and grayscale image data. This framework has the potential to revolutionize infrastructure monitoring applications by enabling real-time, accurate 3D data acquisition for proactive maintenance and informed decision-making.

Proposed Framework

The core of this framework as illustrated in Fig. 2 lies in a multimodal architecture designed to process two data types simultaneously: depth images and corresponding grayscale images. This combined approach allows the model to extract comprehensive information about the fasteners, capturing both visual details and depth. However, a significant challenge arises in acquiring a large and diverse data set of fastener images. Traditional methods often require extensive manual labeling, which can be time-consuming and expensive. To overcome this limitation, we leverage a technique known as self-supervised contrastive learning (Falcon and Cho 2020). This approach enables the model to learn effectively from augmented data, where variations such as rotations and flips are introduced to artificially expand the data set, thereby improving the model's generalization to unseen examples. Self-supervised contrastive learning then directs the model to focus on recognizing similarities and differences between these augmented representations, allowing it to extract meaningful features without extensive manual labeling. Through this enriched data set, the framework achieves real-time, accurate 3D point cloud generation, enhancing infrastructure monitoring systems. The framework's success is largely dependent on a robust data augmentation stage, where techniques such as random rotations, Gaussian noise injection, contrast adjustments, and color jittering are employed. These strategies enrich the training data, improving the model's ability to generalize under varying real-world conditions such as lighting, noise, and object orientations, leading to superior performance in generating 3D point clouds.

This model accomplishes the generation of 3D point clouds solely from 2D image data by capitalizing on several advanced techniques, including self-supervised contrastive learning, Fourier transform-based neural networks (Li et al. 2020), vision transformers (Khan et al. 2022), attention mechanisms (Niu et al. 2021), and Bayesian distribution modeling (Aguilera et al. 2010).

Siamese Network with Enhanced Encoding

The proposed framework employs a Siamese network architecture for representation learning, consisting of two key components: an encoder and a predictor. The encoder encodes information to generate latent representations from grayscale and depth images, which are then processed by the predictor. The predictor functions

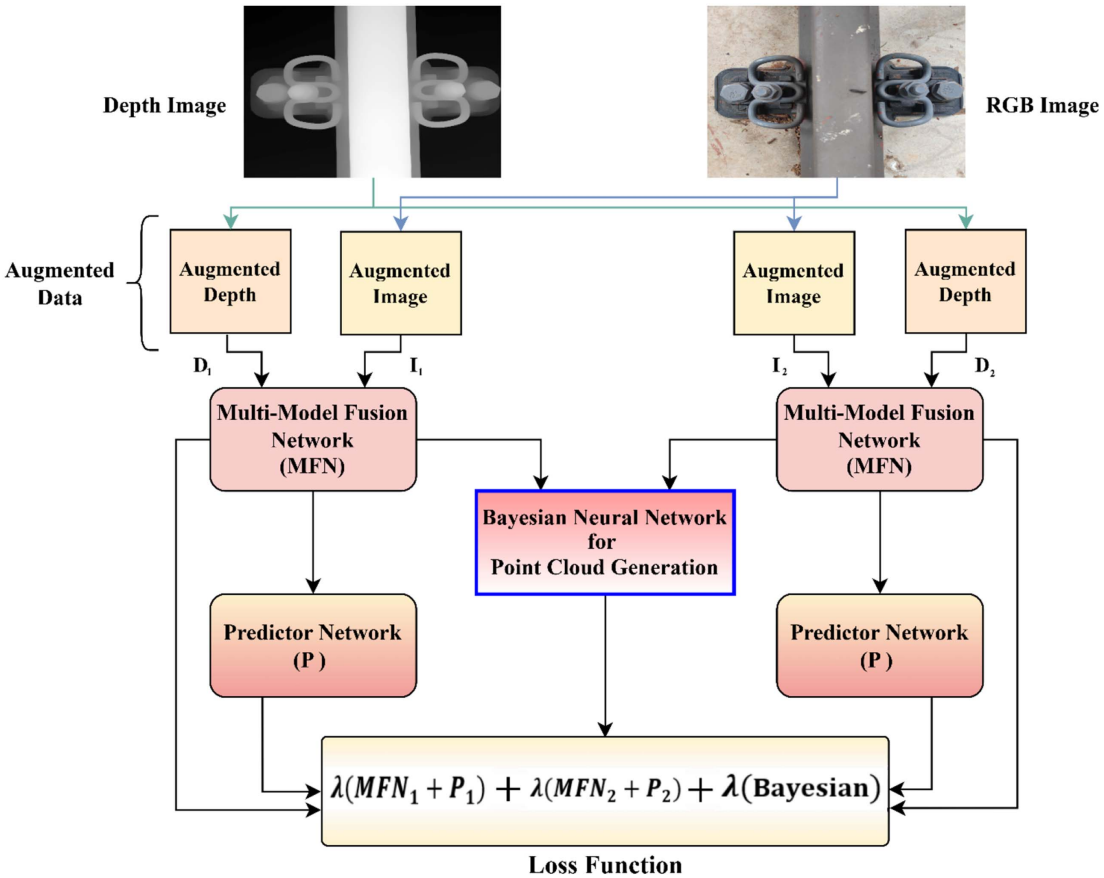


Fig. 2. Proposed framework.

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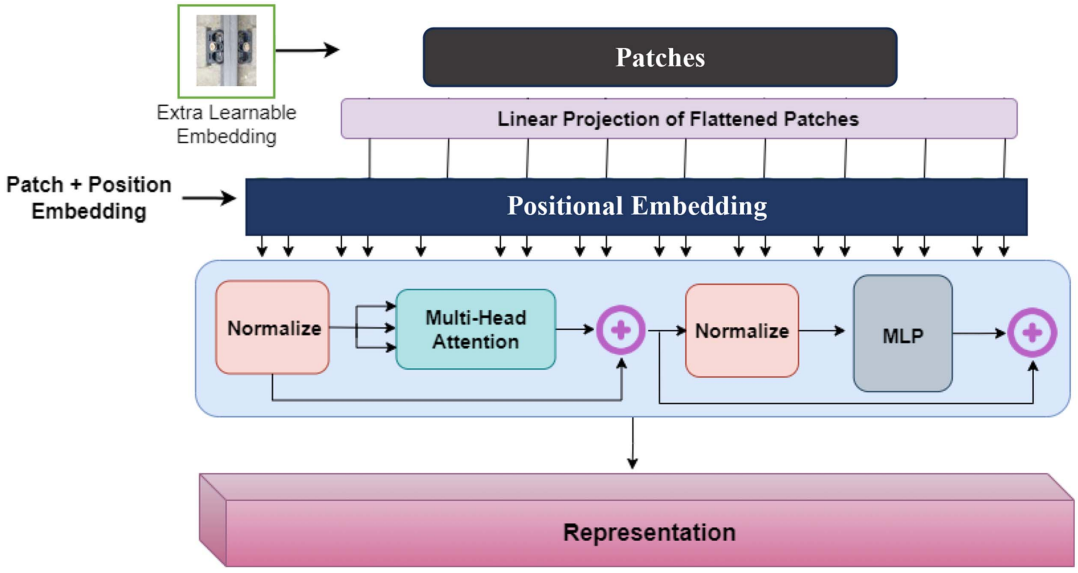


Fig. 3. Vision transformer model.

similarly to a reconstruction algorithm, aiming to maximize the similarity between encoded and predicted representations, thereby training the encoder to produce optimal data encodings. To enhance the encoding process and extract more comprehensive features, the framework integrates a vision transformer, a Fourier network, and a multilevel fusion network (MFN) within the Siamese network architecture, collectively termed the MFN. This architecture efficiently extracts and fuses features from images and depth data, ensuring robust representation learning. Detailed discussions on the vision transformer, Fourier network, multimodal level network (MLN), and overall MFN architecture are provided in subsequent sections.

Vision Transformer. We augment the traditional Siamese network (Guo et al. 2017; Melekhov et al. 2016) by integrating a vision transformer and a Fourier network. Vision transformers (ViTs), as shown in Fig. 3, are a new wave in computer vision, inspired by the success of transformers in natural language processing. Unlike traditional convolutional neural networks (CNNs) that process images pixel by pixel, ViTs divide images into smaller patches. Each patch is embedded into a higher-dimensional space and undergoes positional encoding to preserve its location within the original image. This enables the ViT to grasp the relationships between different parts of the image. The core of the ViT is its transformer encoder, which analyzes the interactions between patches, capturing both local details and global features. This processed information is then utilized for tasks such as image classification or object detection. ViTs offer a robust alternative to CNNs, demonstrating impressive performance across various image analysis benchmarks.

Concurrently, the Fourier network specializes in processing depth images, employing a Fourier transform-based approach to extract depth-specific features. By processing both image and depth data through these specialized networks, our framework achieves a more comprehensive understanding of rail track fasteners.

Fourier Network. The system utilizes a module called the Fourier network (FN) for in-depth analysis of the point cloud data as shown in Fig. 4. While it shares the name with Fourier neural networks used in language processing, this FN focuses on extracting features from the depth information (3D structure) using the Fourier transform (FT). The FT helps the FN identify underlying patterns within the depth data. This FN acts as a feature extractor in two key ways. First, it utilizes various convolution techniques to pull out a rich set of features. These techniques involve breaking down convolutions for efficiency (separable), preserving data at edges (padded), extracting features from individual channels (depth-wise), and capturing spatial relationships (simple). Second, after feature extraction, the FN employs an attention mechanism to filter the features. This mechanism assigns importance weights based on relevance to the task, essentially focusing on the most informative features and discarding less important ones. Finally, the FN uses a technique called "dot attention" to combine all the extracted and filtered features. This creates a single, unified representation that captures important information from both the image and depth data of the point cloud.

Following this initial feature extraction and fusion stage, the framework employs a multilevel fusion network (Fig. 5) to further refine the representation. This network performs a series of fusion steps using dot attention. There are three primary representations obtained at this point: image-based features from the transformer, depth-based features from the Fourier network, and the fused representation generated by the MFN. The multilevel fusion network takes all three of these representations as input and refines them through a series of fusion steps using dot attention. This process ultimately yields a single, unified global representation that captures the most salient features from both the image and depth data. This global representation is then fed into the predictor, which is a residual network comprised of four consecutive transformations. The predictor's role is to refine and reconstruct the global representation. This process yields two sets of representations: one from the encoder and the other from the predictor. The objective is to minimize the distance between these two representations through a loss function. Essentially, the predictor acts as a teacher, guiding the encoder to learn more effective feature representations for point cloud generation.

Multimodal Level Network

The multimodal level network (MLN) plays a critical role in extracting and fusing features from both image and depth data within the enhanced Siamese network of our proposed framework. This network (as shown in Fig. 6) performs two key functions: feature extraction and fusion. Feature extraction is accomplished

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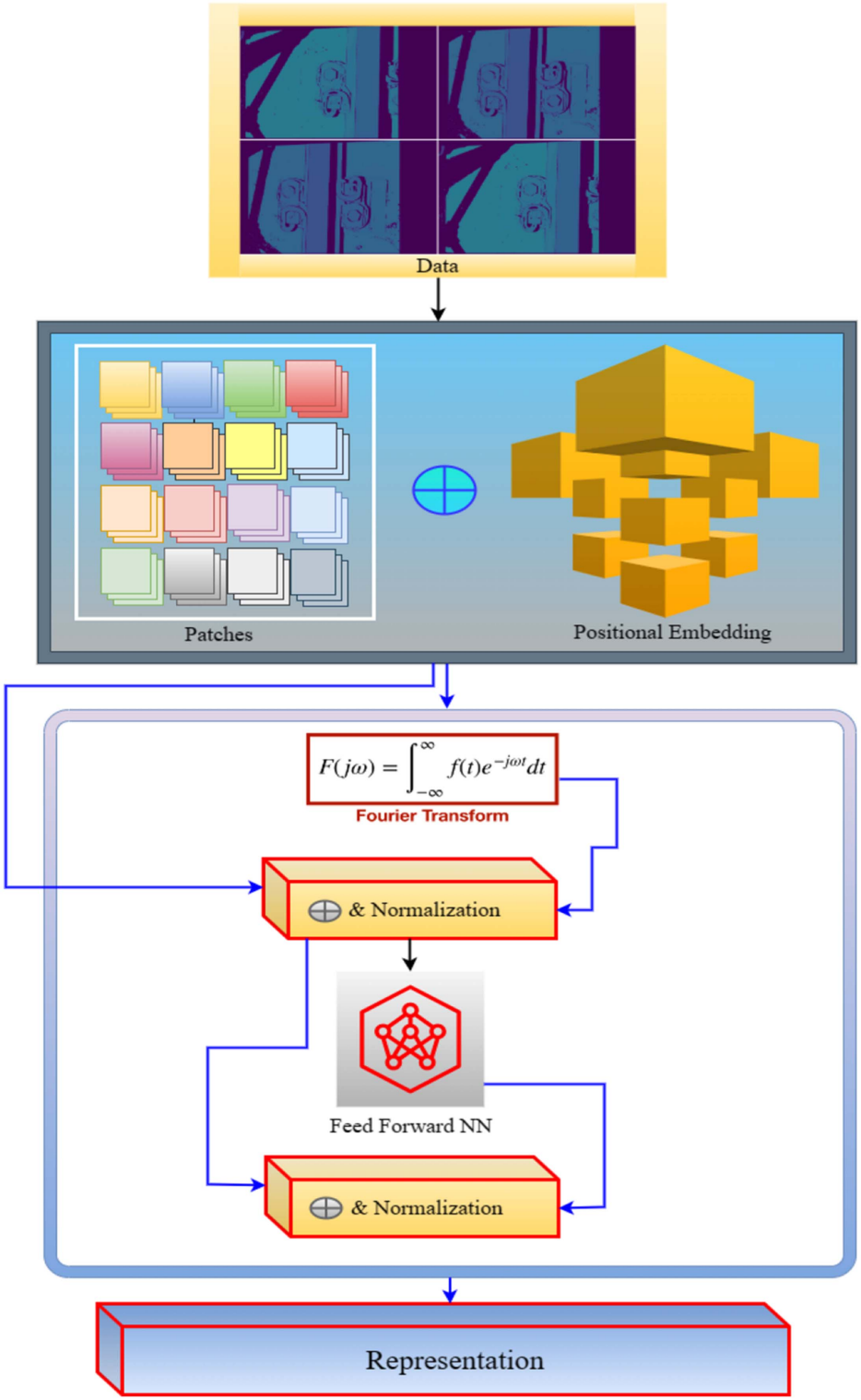


Fig. 4. Fourier neural network.

through various convolution transforms employed by the MLN to capture informative features from the input images and depth maps. These transforms include the following:

- Separable convolutions: This technique breaks down a standard convolution operation into two more efficient steps: depth-wise convolutions followed by a pointwise convolution.
- Padded convolutions: Padding allows the MLN to preserve spatial information during the convolution process, particularly at the edges of the input data. This prevents a loss of valuable information from the borders of the images and depth maps.
- Depth-wise convolutions: These convolutions apply a single filter per input channel, enabling the MLN to extract features specific to each channel independently. This allows for a more granular understanding of the data.
- Simple convolutions: Standard convolutions are also utilized within the MLN to capture more general feature representations

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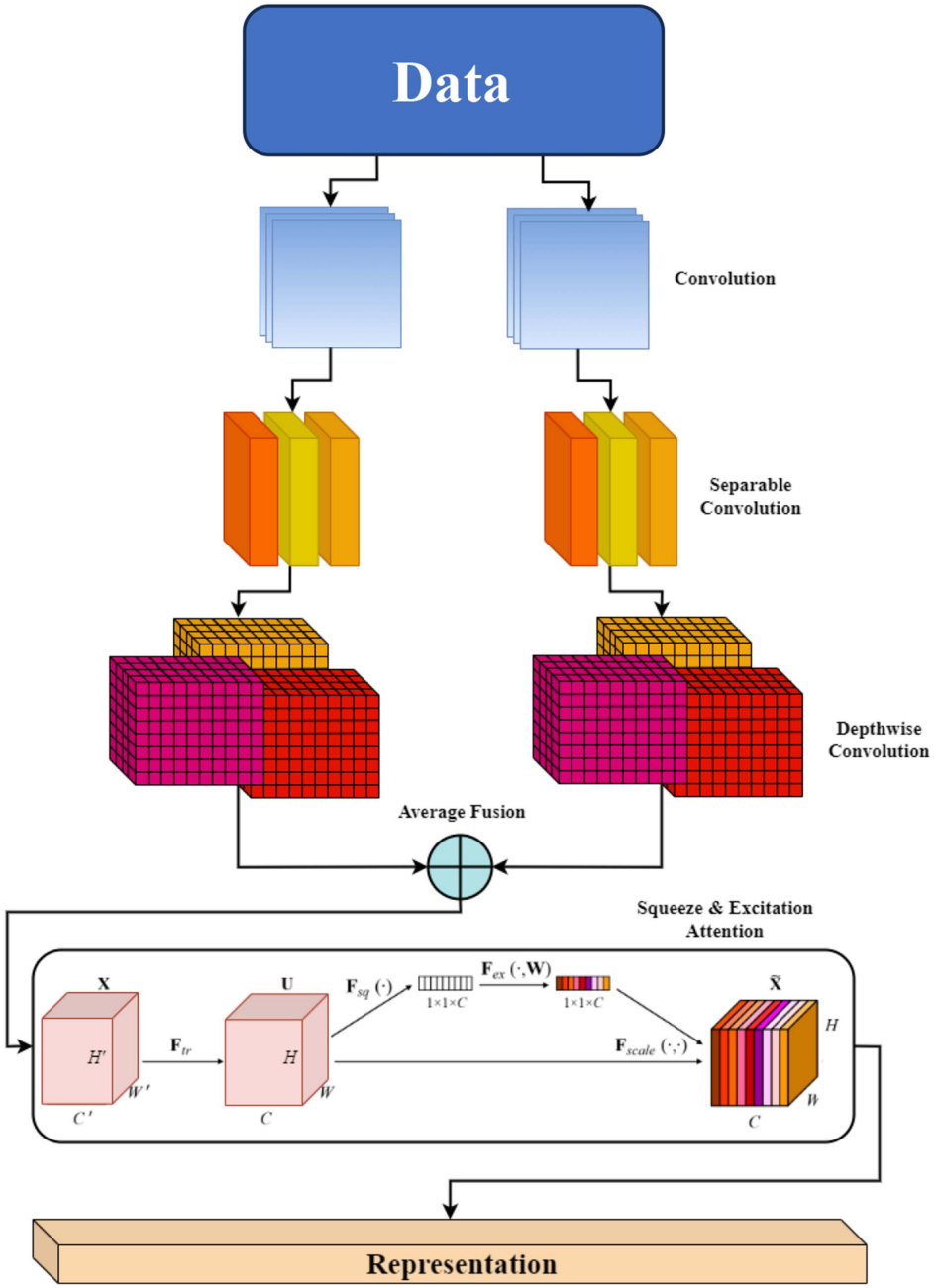


Fig. 5. Multimodal level network (MLN).

from the input data. These convolutions look at broader patterns within the data and complement the channel-specific features extracted by the depth-wise convolutions.

The multilevel fusion network (MLN) utilizes a diverse set of convolution transforms to extract a comprehensive range of features from both image and depth data. These features encompass visual details, depth variations, and spatial relationships within the rail track fasteners. To create a unified representation, the MLN employs a two-stage fusion process:

Squeeze-and-excitation attention mechanism: This mechanism refines the extracted features by assigning importance weights based on their relevance to point cloud generation. It first compresses spatial information into a channel-wise average, allowing the network to evaluate the significance of each feature channel. Fully connected layers then generate scale factors, which are applied to the original feature maps, enhancing the most informative features while suppressing less relevant ones. This process focuses the MLN's attention on the key features necessary for accurate 3D point cloud generation.

Dot attention: After applying the squeeze-and-excitation mechanism, the MLN uses dot attention to further fuse the features. Dot attention calculates compatibility scores between elements in two feature maps, combining them into a single, weighted representation. This fused representation integrates both visual and depth-related information, providing a holistic understanding of the rail track fasteners.

In essence, the MLN acts as a bridge, effectively combining image-centric features from the transformer with depth-specific features from the Fourier network. Through its two-stage attention mechanism, the MLN generates a rich, informative fused

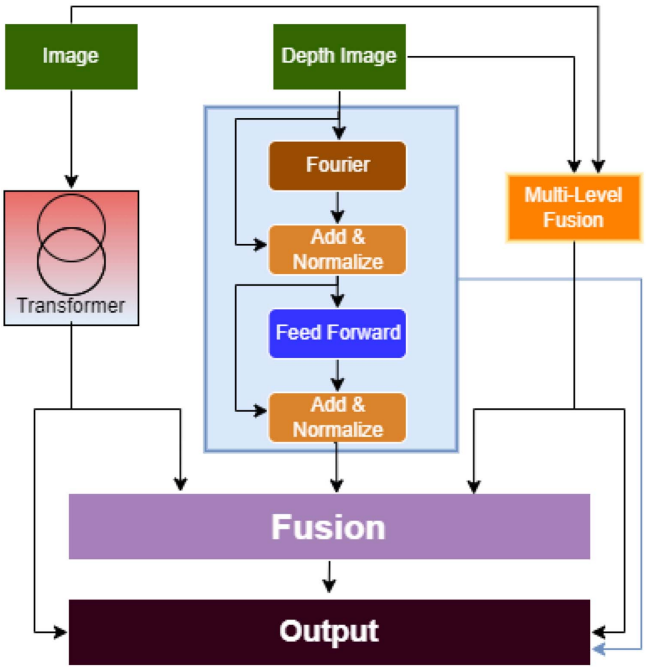


Fig. 6. Multimodal fusion network (MFN) framework.

representation that is critical for producing high-fidelity 3D point clouds in the subsequent stages of the framework. The MFN is a multimodal architecture that consists of a vision transformer, Fourier network, and multilevel fusion network. The process starts with image and depth; firstly images come in a transformer model and the Fourier network is applied to depth images. Another network, called MLN, takes both image and depth and produces mixed feature representation. So overall this MLN is used to learn image representation, depth representation, and mixed representation, which then also fuse to make one final feature representation. This whole network is called a Siamese encoder, which we named MFN because of multimodular processing. These components including the transformer, Fourier network, and MLN are explained in the above sections. Fig. 6 shows the working diagram of MFN. In the next section, we discuss the predictor part. Then these two Siamese encoders (MFN) and predictors are used to learn contrastive representation via self-supervised learning.

Siamese Predictor

The predictor network as shown in Fig. 7 plays a critical role in the Siamese architecture, refining the global representation learned through the multilevel fusion process. This network is a residual network comprised of four consecutive transformation blocks. Each block typically consists of a convolutional layer followed by a batch normalization layer and a rectified linear unit (ReLU) activation function. These residual blocks allow the network to learn complex, nonlinear relationships within the data. The predictor takes the single, unified global representation generated by the multilevel fusion network as input. It then applies the aforementioned transformations within each block to refine and reconstruct this representation. This process essentially attempts to recreate the original input data (both image and depth information) based on the learned global representation. By comparing the reconstructed representation with the original global representation, the framework can assess the effectiveness of the encoding process.

A key objective of the predictor is to minimize the distance between the encoded representation from the encoder (i.e., the global representation before entering the predictor) and the reconstructed representation generated by the predictor itself. This distance minimization is achieved through a loss function. The loss function calculates the difference between the two representations and guides the training process by penalizing the network for significant discrepancies. As the training progresses, the predictor becomes adept at refining the global representation, ensuring that it accurately captures the essential information required for point cloud generation. In simpler terms, the predictor acts as a "teacher" for the encoder within the Siamese network. By striving to reconstruct the global representation, the predictor guides the encoder toward learning more efficient and effective feature representations that are well-suited for the task of generating 3D point clouds from 2D image data.

Bayesian Neural Network for Point Cloud Generation

The ultimate goal of the proposed framework, as shown in Fig. 8, is to generate 3D point clouds of rail track fasteners based on the learned representations. This section delves into the intricacies of the Bayesian neural network employed for this crucial task. The core idea behind this approach lies in acknowledging the inherent uncertainty associated with the point cloud generation process. Real-world data often exhibit variations and noise, and the mapping from 2D representations to a 3D point cloud is inherently complex and may not have a single, definitive solution. To address this uncertainty, we leverage a Bayesian learning framework.

Our framework utilizes four distinct representations as input for the Bayesian neural network:

Image Representation: This representation captures visual details extracted from the image data by the transformer network.

Depth Representation: This representation encapsulates depth-related information obtained from the depth maps through the Fourier network.

Combined Representation (MLN): This representation integrates features from both image and depth domains, generated by the multilevel fusion network (MLN).

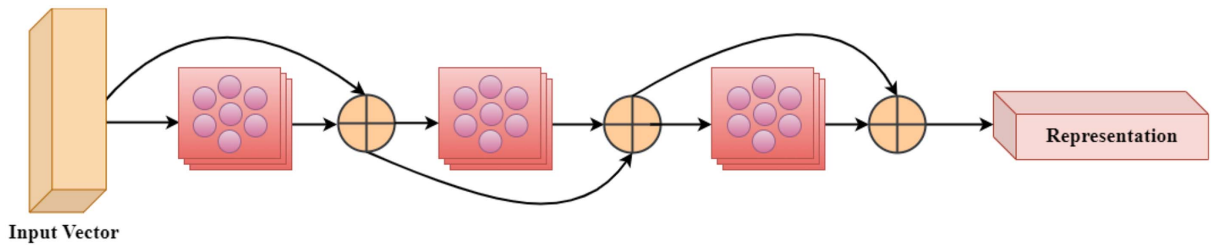


Fig. 7. Predictor network.

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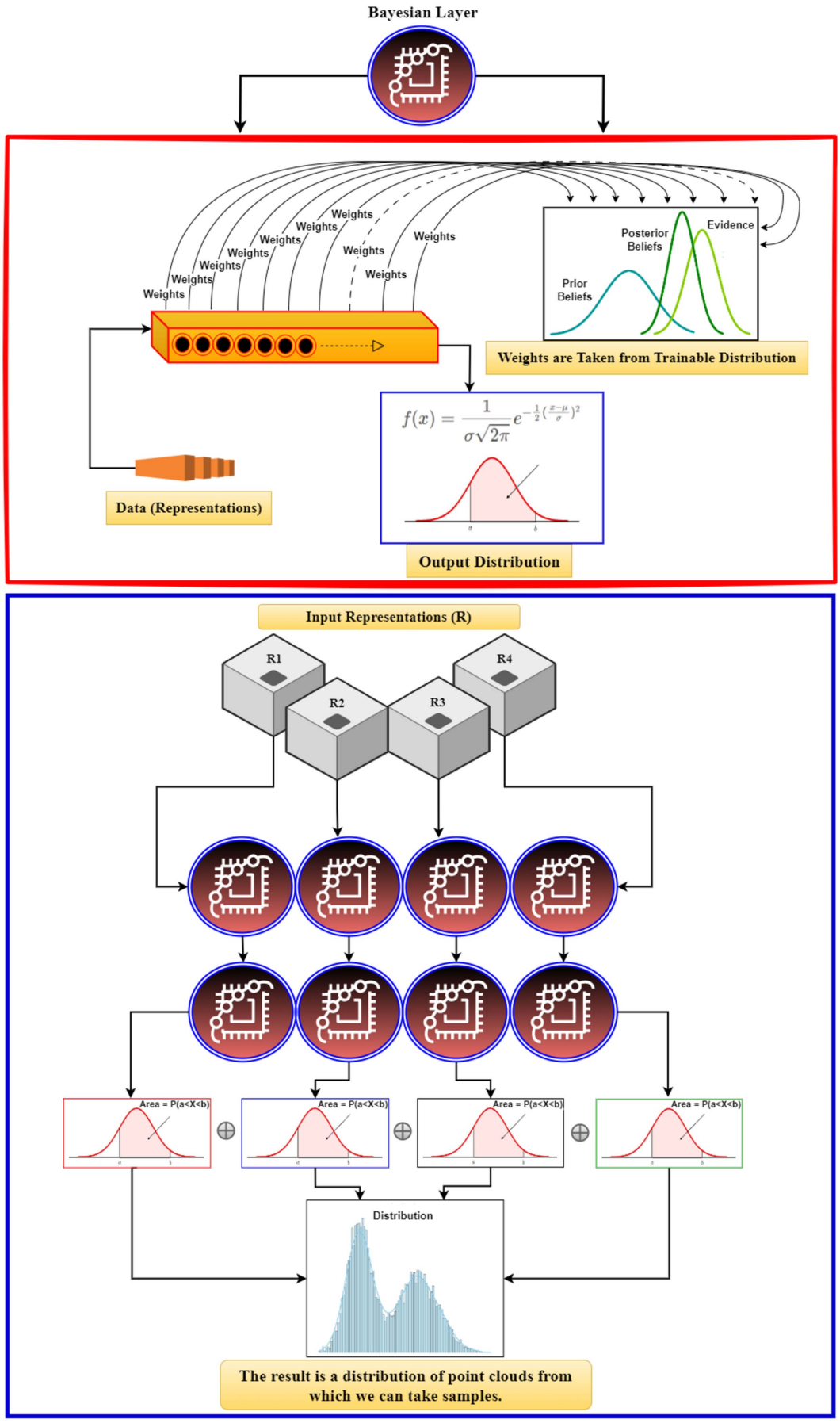


Fig. 8. Bayesian neural network for point cloud generation.

Global Combined Representation: This representation is the culmination of the multimodal fusion process, incorporating information from all three aforementioned representations.

Each of these representations offers unique knowledge about the rail track fasteners:

- The image representation provides details regarding the visual appearance of the fasteners.
- The depth representation captures information about the spatial distribution of depth values.
- The combined representation (MLN) offers a more holistic understanding by combining visual and depth cues.
- The global combined representation acts as a comprehensive summary, leveraging information from all available sources.

To account for the inherent uncertainty in point cloud generation, the framework employs a Bayesian approach. Instead of predicting a single, deterministic point cloud, the Bayesian neural network learns the probability distribution underlying the point cloud generation process. This distribution captures the range of possible point clouds that could be generated based on the input representations.

Bayesian Small Networks: We utilize separate small neural networks for each of the four input representations. These networks are tasked with learning the probability distribution of the point cloud for each representation. This allows the network to capture the unique contribution of each representation to the overall point cloud generation process.

Distribution Fusion: The individual probability distributions learned from each small network are not directly used for point cloud generation. Instead, a complex, nonlinear fusion process combines these distributions to create a single, unified global distribution that encapsulates the combined knowledge from all four representations. This fusion step leverages techniques such as deep amortization, allowing the framework to learn a more comprehensive and informative distribution for generating high-fidelity 3D point clouds.

Point Cloud Sampling: Once the global distribution is established, the framework utilizes a sampling technique to generate a point cloud. This sampling process essentially draws a sample from the learned distribution. Since the distribution captures the inherent uncertainty in the data, multiple samples can be drawn, resulting in a set of potential point clouds that are all plausible given the input representations. This ability to generate multiple point cloud samples reflects the inherent variability present in real-world data.

By leveraging a Bayesian framework (as illustrated in Fig. 8) and sampling from the learned distribution, the proposed approach effectively manages uncertainty in point cloud generation. This probabilistic method offers a nuanced understanding of potential variations in the generated point clouds, providing valuable insights for infrastructure monitoring. Rather than producing a single "best guess" point cloud, the framework generates multiple potential point clouds that represent the range of possibilities based on the input data and associated uncertainties. This capability is particularly beneficial for tasks such as anomaly detection in rail track maintenance. By comparing generated point clouds to a reference or identifying significant deviations among samples, the framework can reveal potential issues with rail track fasteners, facilitating proactive maintenance strategies.

Results of Simulation

In this section, an examination of the system's performance on the rail track fasteners data set is undertaken. The discussion encompasses an elucidation of the data set, delineation of data augmentation techniques, and explication of data preprocessing methods. Subsequently, a comprehensive analysis of the proposed system is conducted, focusing on representation learning and point cloud generation. Following this, a complexity analysis is executed to assess the intricacies of the system, given its composition of multiple algorithms. This process is conducted for training the model by utilizing Google Colab with a T4 GPU, optimizing our model with specific hyper parameters: a batch size of 1, 30 epochs, and a learning rate of 0.001. For the evaluation of the generated point clouds, we employed three distinct metrics: Chamfer distance, Euclidean contrastive metric, and Hausdorff distance, ensuring a comprehensive assessment of the point cloud quality and alignment.

Data Set and Preprocessing

This research used a unique data set of rail fasteners captured on a high-speed railway track. The data consists of rail fasteners' grayscale and depth images captured on a high-speed railway track. A Deep Vision Camera SSZN-SR5320, known for its structured light data acquisition capabilities, was mounted on a comprehensive track inspection vehicle as shown in Fig. 9. To ensure high-quality data, the vehicle traveled at a controlled speed of 10 km/h. Four strategically positioned sets of joint line-structured light cameras scanned the track. These cameras, mounted 500 mm above the track, offered a combined scanning width of 2,400 mm, effectively capturing the entire area of interest. The data collection focused on two specific track slabs, resulting in a high-density point cloud containing approximately 50 million data points per slab. These point clouds boast impressive resolutions of 50 μm × 50 μm horizontally and 7.5 μm vertically. Additionally, the captured 2D grayscale images had a resolution of 1,920 × 1,200 pixels, while the depth images offered the same resolution with an additional channel containing further information. This unique data collection process ensured the acquisition of rich and detailed information about the rail fasteners, providing a valuable foundation for the deep-learning framework.

To train the model efficiently, the high-resolution grayscale (1,920 × 1,200 pixels) and depth images (1,920 × 1,200 pixels with three channels) were resized to a standard 128 × 128 pixels. Additionally, pixel values were normalized using min–max scaling

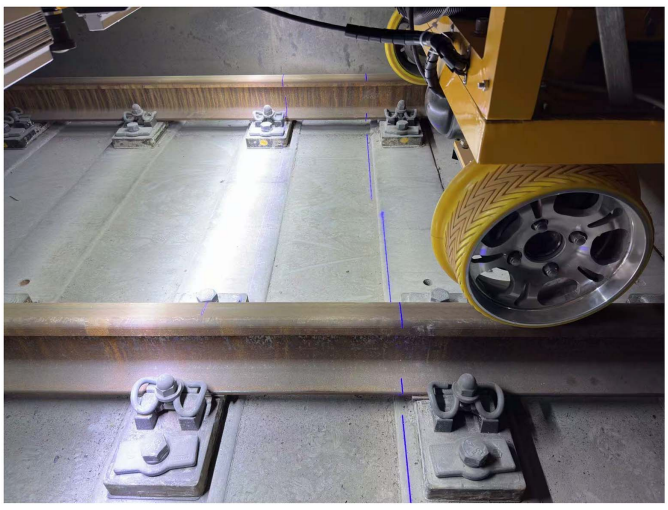


Fig. 9. Comprehensive track inspection vehicle.

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Fig. 10. 3D point cloud of rail track fasteners data set.

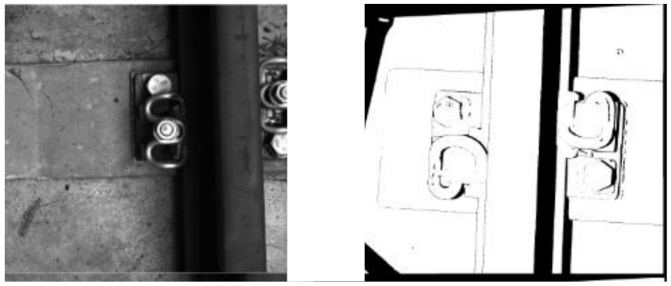


Fig. 11. Rail tract fasteners data set (first column shows the grayscale images, second column shows the depth images).

to prevent specific features from biasing the training process. Finally, both image types were converted into a format compatible with the deep-learning model's pipeline. The evaluation process utilizes a custom-collected data set specifically designed for rail fastener analysis. While the data set (samples shown in Figs. 10 and 11) comprises 30 corresponding depth and grayscale images for each fastener type, this limited quantity could potentially impact the model's performance and efficiency when trained with traditional supervised learning methods. To overcome this challenge, the framework leverages self-supervised contrastive learning. This approach enables the model to effectively extract valuable representations from the available data, even with a relatively small data set. By learning from artificially augmented versions of the existing data and focusing on identifying similarities and differences within these variations, the model can learn meaningful features without requiring a vast amount of manually labeled data. These preprocessing steps were crucial for ensuring successful training and evaluation of the point cloud generation model. Downsizing the high-density point clouds and normalizing the data streamlined the processing workflow and promoted stable training. Ultimately, these measures contributed to the development of a more robust and reliable model capable of generating highly accurate 3D representations of rail track fasteners.

Data Augmentation

To enrich the training data and improve the model's generalizability, various data augmentation techniques were employed on data as shown in Fig. 12. These included random rotations to enhance object orientation invariance, Gaussian noise injection to increase robustness to noise, random contrast adjustments to enable generalizability across diverse lighting conditions, and color jittering to promote robustness to variations in color distribution. By leveraging these techniques, the framework benefits from a richer and more diverse training set, leading to enhanced performance and improved generalizability of the generated 3D point clouds of rail fasteners under diverse real-world conditions.

Performance Metrics

Evaluating the performance of the deep-learning framework for point cloud generation requires a set of well-defined metrics that quantify various aspects of the generated point clouds and the associated representation learning processes. Choosing the appropriate metrics allows for a comprehensive understanding of the framework's strengths and weaknesses.

Chamfer Distance

This metric quantifies the dissimilarity between two sets of points in 3D space. It measures the average distance from each point in one set to its nearest neighbor in the other set, and vice versa. Lower chamfer distance values indicate a better match between the generated point clouds and the target point clouds representing the actual rail track fasteners. A significant difference between the chamfer distance for different data augmentation approaches can highlight the strategies that lead to more accurate point cloud generation.

The Chamfer distance (CD) between two-point clouds, P and Q (with P having n points and Q having m points), is defined by the equation

$$CD(P,Q) = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{min}_{j=1}^{m} \left\| \boldsymbol{p}_i - \boldsymbol{q}_j \right\|^2 + \frac{1}{m}\sum_{j=1}^{m} \boldsymbol{min}_{i=1}^{n} \left\| \boldsymbol{q}_j - \boldsymbol{p}_i \right\|^2 \tag{1}$$

where $\boldsymbol{p}_i$ and $\boldsymbol{q}_j$ = points in P and Q, respectively. The resulting Chamfer distance values, ranging from 0.3926 to 0.4050, indicate dissimilarity, with lower values being more desirable as they suggest a better match between the generated and target point clouds. Notably, noise augmentation achieves the lowest Chamfer distance, underscoring its contribution to more accurate point cloud generation.

Hausdorff Distance

This metric complements the Chamfer distance by measuring the maximum distance between two sets of points in 3D space. It identifies the point in one set that has the greatest distance to its nearest neighbor in the other set. Similar to the Chamfer distance, lower Hausdorff distance values indicate a closer match between the generated and target point clouds. Analyzing the Hausdorff distance for different augmentation strategies can provide insights into the overall alignment and shape fidelity of the generated point

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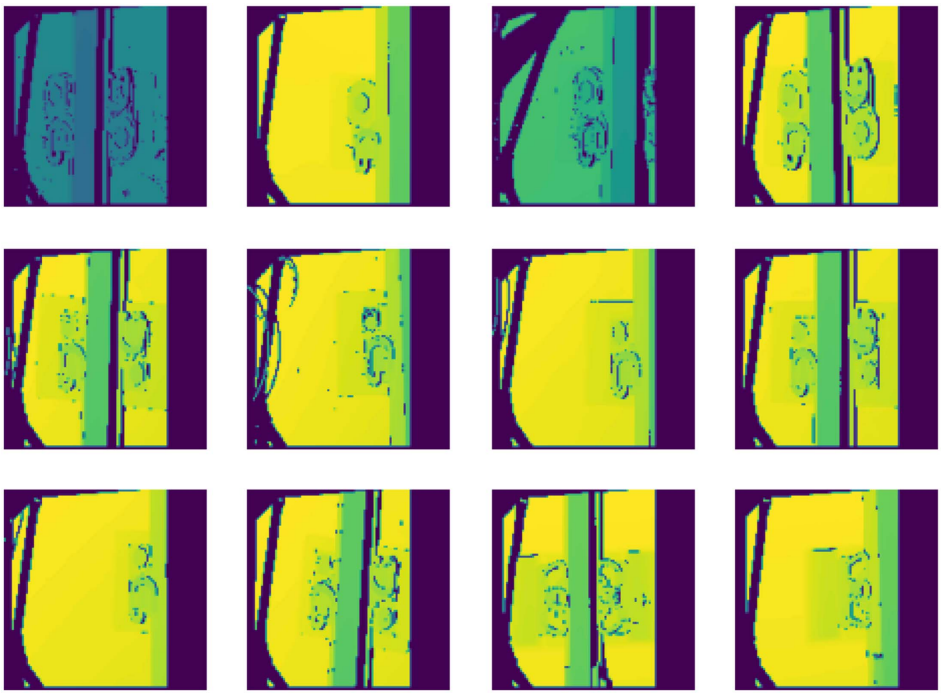


Fig. 12. Images and depth after augmentation.

clouds. The directed Hausdorff distance ($HD(\boldsymbol{P},\boldsymbol{Q})_{\boldsymbol{directed}}$) from point cloud P to point cloud Q is expressed mathematically as

$$HD(\boldsymbol{P},\boldsymbol{Q})_{\boldsymbol{directed}} = \boldsymbol{max}_{\boldsymbol{p}\in\boldsymbol{P}}(\boldsymbol{min}_{\boldsymbol{q}\in\boldsymbol{Q}}\|\boldsymbol{p}_{\boldsymbol{i}} - \boldsymbol{q}_{\boldsymbol{j}}\|) \tag{2}$$

This equation signifies that, for each point $\boldsymbol{p}$ in point cloud P, the distance to its nearest neighbor in point cloud Q is determined. The maximum of these minimum distances across all points in P characterizes the directed Hausdorff distance.

To obtain the overall undirected Hausdorff distance ($HD(\boldsymbol{P},\boldsymbol{Q})_{\boldsymbol{undirected}}$) between the two point clouds, both [HD(P, Q)] and [HD(Q, P)] are calculated. The undirected Hausdorff distance is then determined as the maximum of these two values:

$$HD(\boldsymbol{P},\boldsymbol{Q})_{\boldsymbol{undirected}} = \mathbf{max}(\boldsymbol{max}_{\boldsymbol{p}\in\boldsymbol{P}}(\boldsymbol{min}_{\boldsymbol{q}\in\boldsymbol{Q}}\|\boldsymbol{p} - \boldsymbol{q}\|), \boldsymbol{max}_{\boldsymbol{q}\in\boldsymbol{Q}}(\boldsymbol{min}_{\boldsymbol{p}\in\boldsymbol{P}}\|\boldsymbol{q} - \boldsymbol{p}\|)) \tag{3}$$

Now, considering the results, the computed Hausdorff distance values range from 1,063.79 to 1,140.1226. In the context of this evaluation, lower values are desirable as they indicate a closer match between the two-point clouds. The application of noise augmentation, along with contrast or color jitter, results in relatively lower Hausdorff distance values, suggesting improved alignment. The consistent effectiveness of noise augmentation across multiple metrics, including Chamfer distance and representation learning, underscores its role in achieving accurate point cloud generation and facilitating effective representation learning.

Euclidean Contrastive Loss

The Euclidean contrastive loss metric is crucial during neural network training as it serves as a loss function guiding the network to learn robust data representations. This metric measures the Euclidean distance between pairs of samples in the training data, aiming to cluster similar samples together in the embedding space while separating dissimilar ones. Minimizing the Euclidean contrastive loss encourages the network to extract features that effectively distinguish between different types of rail track fasteners or variations within the same type. Lower values of this loss indicate more effective representation learning. Analyzing this metric across various data augmentation techniques can identify which strategies enhance the model's ability to develop discriminative features for accurate point cloud generation.

The Euclidean contrastive loss is typically formulated for a pair of examples (a, b) with a binary label indicating whether they are similar ($y = 1$) or dissimilar ($y = 0$). The loss function is expressed as follows:

$$L(a,b,y) = \frac{(\mathbf{1}-\boldsymbol{y})}{\mathbf{2}}\|\boldsymbol{a}-\boldsymbol{b}\|^2 + \frac{\boldsymbol{y}}{2}\boldsymbol{max}(\mathbf{0},\boldsymbol{m}-\|\boldsymbol{a}-\boldsymbol{b}\|)^2 \tag{4}$$

where $\boldsymbol{m}$ is a margin parameter that sets the threshold for dissimilarity. The computed Euclidean contrastive loss values in this context range from 0.3142 to 0.3327. Lower values are favored as they signify more effective representation learning within the neural network. Notably, noise augmentation consistently yields the lowest Euclidean contrastive loss, indicating its positive impact on improving the quality of learned representations during the training process.

To rigorously evaluate the accuracy of our reconstructed models, we utilized three complementary metrics: Chamfer distance, Hausdorff distance, and Euclidean contrastive loss. Chamfer distance provides a thorough measure of the average bidirectional discrepancy between the reconstructed model and the ground truth, highlighting local deviations and fine detail discrepancies. Hausdorff distance complements this by focusing on the maximum deviation between points in the reconstructed and ground truth models, identifying extreme discrepancies and outliers. Euclidean contrastive loss assesses the quality of learned latent representations, ensuring that similar 3D inputs are represented by closely aligned latent codes. This loss function guides the model in learning an effective feature space where objects with similar shapes are positioned near each other, even if their poses or viewpoints vary. Together, these metrics offer a comprehensive evaluation of both geometric accuracy and the quality of latent feature representations,

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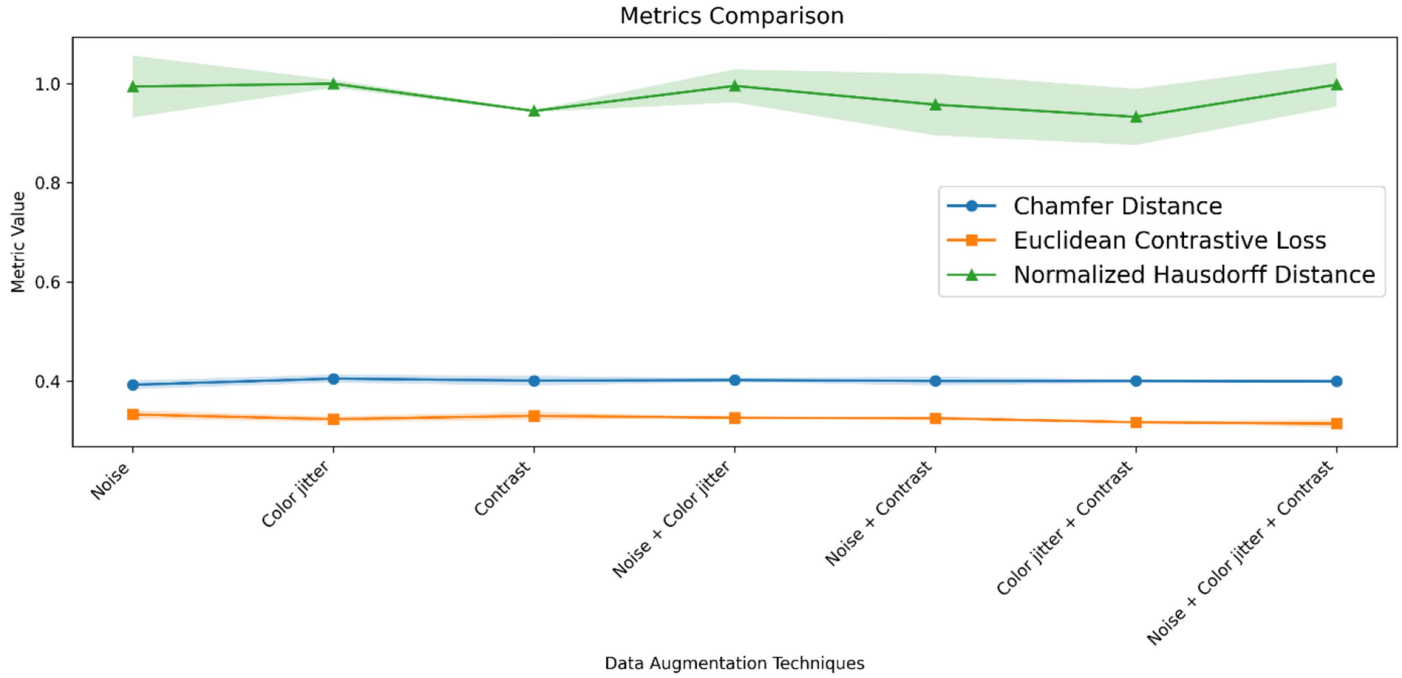


Fig. 13. Performance of data augmentation techniques.

enabling us to detect both global and local reconstruction discrepancies and ensure the model captures the essential features of the 3D objects effectively.

Fig. 13 visually encapsulates the quantitative outcomes of our experimentation, providing a comprehensive illustration of the performance of various data augmentation techniques on Chamfer distance (CD), Euclidean contrastive loss (L), and Hausdorff distance for point clouds (HD). The plot serves as a graphical representation of the nuanced impacts of each augmentation strategy and their combinations on the dissimilarity metrics. Observing the trends, it becomes evident that noise augmentation consistently leads to the most favorable outcomes across CD and L metrics, aligning with the numeric results presented in the table. The curves associated with noise augmentation exhibit the lowest points, emphasizing its efficacy in achieving accurate point cloud generation and improving the quality of learned representations. Moreover, this allows for a comparative analysis of the augmentation techniques and their combinations. It provides insights into how the joint application of noise, color jitter, and contrast augmentation influences the dissimilarity metrics. The curves for this combined strategy, while demonstrating slight fluctuations, consistently showcase competitive performance, reinforcing the potential synergy among diverse augmentation methods. In essence, Fig. 8 acts as a visual aid to complement the tabulated results, offering a holistic view of the relative impacts of different augmentation strategies on the dissimilarity metrics and further aiding in the interpretation of the experimental findings.

In our evaluation process, various data augmentation strategies were applied individually and in combination to assess their impact on the quality of point cloud generation, as shown in Table 1. The Chamfer distance (CD) metric, indicative of dissimilarity between generated and target point clouds, showcased nuanced responses to different augmentation techniques. Notably, the introduction of noise augmentation resulted in the lowest CD value at 0.3926, suggesting a more accurate match between the generated and target point clouds. The Euclidean contrastive loss (L), a crucial component in training neural networks for metric learning, exhibited a consistent trend. Once again, noise augmentation emerged as particularly effective, yielding the lowest loss value at 0.3142 and emphasizing its contribution to enhancing the learned representations. Simultaneously, the Hausdorff distance for point clouds (HD) revealed distinctive patterns in response to augmentation strategies. The combination of noise, color jitter, and contrast augmentation produced the lowest HD value at 1,063.79, indicating superior alignment between the point clouds. These findings underscore the nuanced and strategy-dependent impacts of data augmentation on the quality of generated point clouds and the associated representation learning processes.

In Table 1, it is evident that noise-based data enhancement achieves the lowest Chamfer distance (0.3926) among all tested

Table 1. Detailed evaluation data augmentation techniques

No.	Data augmentation (only)	Chamfer distance for PC	Euclidean contrastive loss	Hausdorff distance for PC
1	Noise	0.3926	0.3327	1,133.550
2	Color jitter	0.4050	0.3232	1,140.1226
3	Contrast	0.4010	0.3300	1,077.2986
4	Noise + Color jitter	0.4021	0.3261	1,135.091
5	Noise + Contrast	0.4003	0.3251	1,091.942
6	Color jitter + Contrast	0.4003	0.3171	1,063.79
7	Noise + Color jitter + Contrast	0.3996	0.3142	1,137.9042

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augmentation techniques. This finding highlights the unique advantages of noise enhancement in promoting more accurate point cloud generation. The introduction of noise during training enhances the model's robustness and generalization capabilities by exposing it to a wider range of variability, thus preventing overfitting and improving performance on noisy real-world data. A detailed comparison is provided in Fig. 14.

To understand why noise-based data enhancement achieves notable improvements and its distinct advantages over other data augmentation methods, it is crucial to explore the nature of noise and its effects on model training. Noise, when applied judiciously, introduces variability that enhances the model's robustness, enabling it to better generalize to real-world data which often contains inherent noise. This exposure to noisy data during training helps the model mitigate noise effects, improving its performance and leading to a more accurate representation of the underlying data structure, as evidenced by reduced Chamfer distances. Unlike other augmentation techniques, such as color jitter and contrast adjustment, which mainly alter visual appearance without incorporating the same level of variability, noise augmentation encourages the model to learn generalized features, thereby reducing overfitting.

Table 1 highlights that while noise augmentation alone achieves a significant reduction in Chamfer distance, indicating enhanced model robustness, it does not yield the lowest Euclidean contrastive loss (L) values. The optimal L values are observed with combined augmentation techniques, specifically noise combined with color jitter and contrast, achieving the lowest L index scores (0.3171 and 0.3142, respectively). This indicates that although noise augmentation significantly improves model performance, the most substantial gains are realized when it is used in conjunction with other augmentation methods. The synergistic effect of combining noise with color jitter and contrast underscores the importance of a holistic augmentation approach to maximize model accuracy and robustness. Thus, while noise augmentation contributes notably to performance improvements, it is the integration of multiple augmentation techniques that leads to the most effective enhancement in model performance, demonstrating that combined augmentations are more beneficial than any single isolation method.

Table 1 reveals that the lowest Hausdorff distance (HD) value of 1,063.79 is achieved with the combination of color jitter and contrast augmentations, whereas the addition of noise leads to a slight increase in HD. It is crucial to recognize that our study aims to optimize multiple metrics, including Chamfer distance, Euclidean contrastive loss (L), and Hausdorff distance (HD), simultaneously. Although the combination of color jitter and contrast achieves the lowest HD, this does not necessarily indicate overall superiority in point cloud quality. Our findings demonstrate that noise augmentation significantly reduces Chamfer distance and Euclidean contrastive loss, which are also critical for assessing point cloud quality. Therefore, despite the minor increase in HD, noise augmentation substantially improves other quality measures. This highlights the trade-offs and synergistic effects involved in optimizing these metrics. Noise augmentation, while slightly increasing HD, enhances Chamfer distance and Euclidean contrastive loss, indicating an overall improvement in point cloud generation quality when evaluated across multiple metrics. Additionally, a holistic evaluation approach, rather than focusing solely on HD, provides a balanced assessment of augmentation strategies. As shown in Table 1, noise augmentation achieves the lowest Chamfer distance of 0.3926 and one of the lowest Euclidean contrastive loss values at 0.3327. Although the HD slightly rises to 1,133.550, the overall improvements in Chamfer distance and Euclidean contrastive loss justify this minor increase. Complementary to these quantitative metrics, visualization techniques can provide further insights. Plotting the impact of various augmentation strategies on Chamfer distance, Euclidean contrastive loss, and Hausdorff distance enables a comparative analysis of the effects of each augmentation approach and its combinations on point cloud quality.

Metrics with Uncertainties

In this study, Fig. 15 serves as a comprehensive visual exploration of the impact of various data augmentation techniques on three distinct metrics—Chamfer distance, Euclidean contrastive loss, and Hausdorff distance. The figure is organized into rows, each dedicated to a specific data augmentation method, including noise, color jitter, contrast, and combinations thereof.

The left-side bar charts offer a categorical overview of metric values, providing a concise snapshot of the differential effects of each augmentation method on the metrics. Notably, noise appears to exert the least influence on the metrics, while combinations such as noise, color jitter, and contrast exhibit discernible variations. This categorical representation facilitates a quick assessment of the relative impact of different augmentation techniques. On the right side of the figure, line plots delve deeper into individual metrics, focusing on mean values. The shaded regions surrounding each line convey the associated uncertainty bounds, acknowledging potential variations attributable to measurement uncertainties or other factors. Each metric is distinguished by color, enabling a clear differentiation of their respective contributions. This detailed graphical representation affords a nuanced understanding of how each data augmentation method influences the metrics and provides valuable insights into the potential uncertainties associated with these effects.

This study explored various data augmentation combinations to understand their effects on point cloud quality comprehensively. Each augmentation method, whether applied individually or in combination, demonstrated unique impacts on metrics such as Chamfer distance, Euclidean contrastive loss, and Hausdorff distance. These impacts provide valuable insights into how different augmentations contribute to overall point cloud quality. Following this, it primarily utilized the combination of noise, color jitter, and contrast augmentations. This combination was chosen based on its ability to provide a balanced improvement across all evaluated metrics. While it is true that the combination of color jitter and contrast alone achieved the lowest Hausdorff distance, the inclusion of noise significantly reduced Chamfer distance and Euclidean contrastive loss. This holistic approach to evaluation ensured that we did not focus solely on optimizing a single metric but rather aimed for a comprehensive enhancement of point cloud quality. Therefore, for future research, we recommend employing the combination of noise, color jitter, and contrast augmentations. This approach offers a balanced and robust enhancement of point cloud quality, addressing various aspects of the data simultaneously.

Segmentation and Surface Reconstruction of 3D Point Clouds

The ability to analyze and understand 3D structures plays a crucial role in various disciplines. This document delves into a methodology for extracting meaningful information from 3D point clouds, which are essentially collections of data points representing objects or features in space. These points, typically defined by X, Y, and Z coordinates, capture the spatial distribution of the elements within the scene. Here, we explore a two-pronged approach: segmentation and surface reconstruction, to unlock the hidden details within these point clouds.

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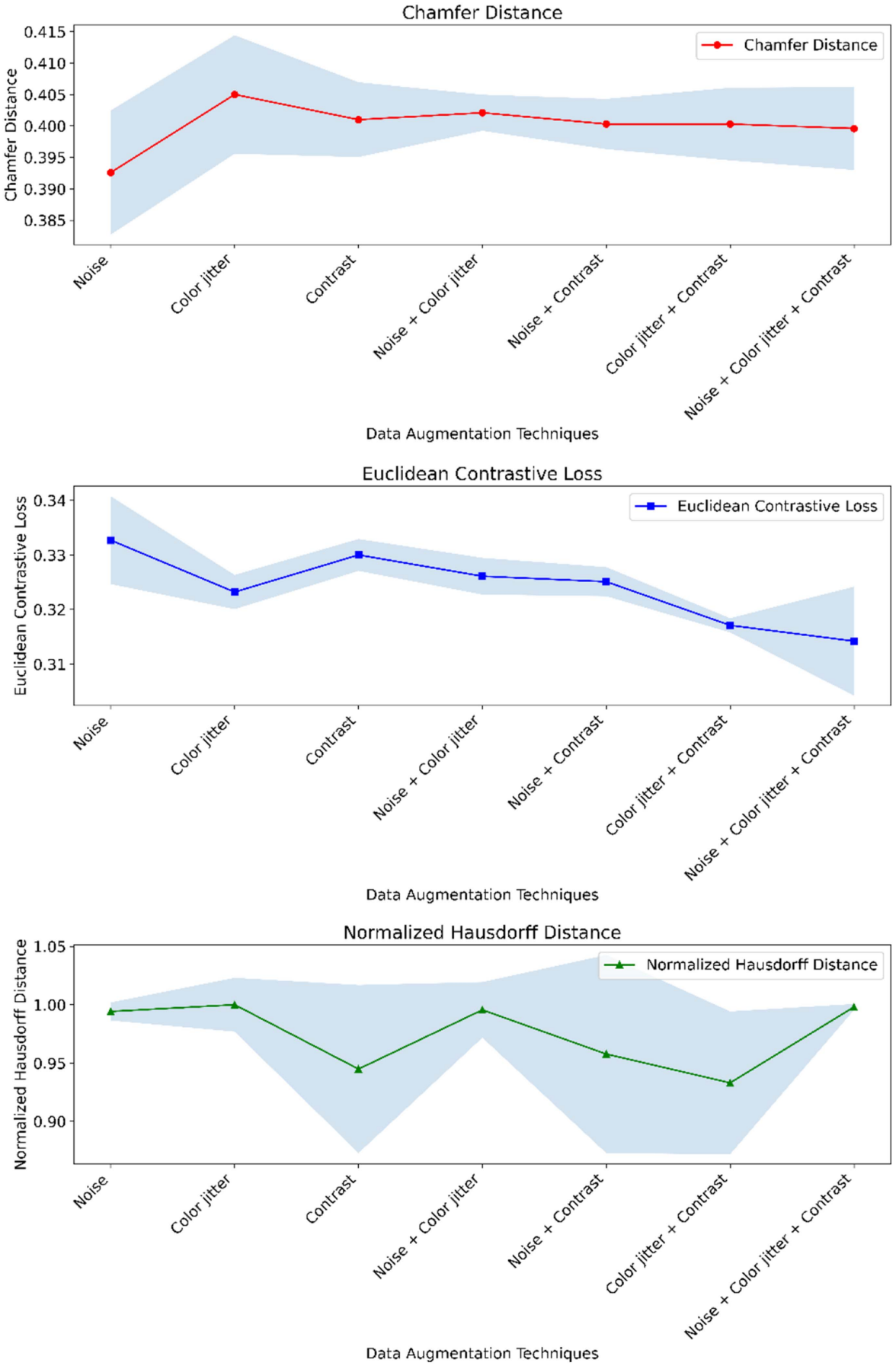


Fig. 14. Detailed comparison of data augmentation techniques.

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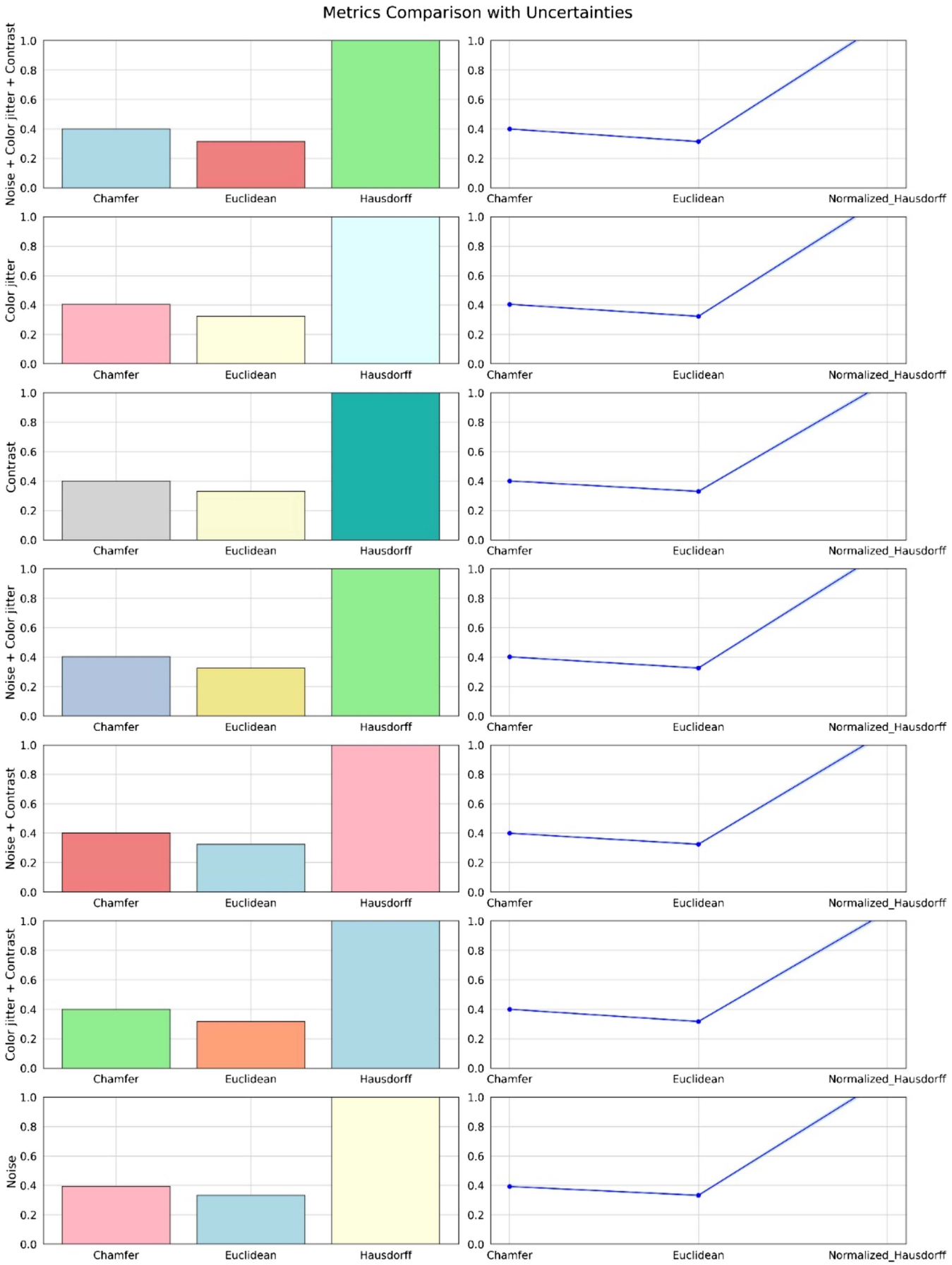


Fig. 15. Impact of data augmentation techniques.

Segmentation

The initial step, segmentation, focuses on isolating specific features of interest from the overall point cloud. Imagine a point cloud representing a bridge with fasteners attached. Segmentation allows us to separate the bridge, the main plane structure, from the individual fasteners such as bolts or screws. This isolation facilitates the individual characterization and analysis of each component. The Random Sample Consensus (RANSAC) algorithm serves as a powerful tool for achieving this segmentation objective (Li et al. 2017). RANSAC is adept at identifying outliers within a point cloud. In our bridge example, the bridge itself represents the dominant plane. RANSAC effectively identifies points deviating from this dominant plane, effectively pinpointing the location of the fasteners. Fig. 16 visually depicts the segmentation process. The first column showcases the raw point cloud encompassing both the bridge and the fasteners. The second column presents the isolated bridge region identified as an outlier by RANSAC. Finally, the third column displays the extracted fastener, now separated from the bridge structure.

Having isolated the desired features, we can delve into the second step: surface reconstruction. This process aims to create a more detailed and visually informative representation of the spatial relationships within the point cloud. Here, cubic interpolation comes into play. Interpolation essentially estimates missing data points based on the surrounding known values. In the context of point clouds, this translates to estimating Z values for each point, effectively constructing a continuous surface that reflects the underlying 3D structure.

The X, Y, and Z coordinates of the original data points provide the foundation for this interpolation. By estimating missing Z values, we create a smooth surface that bridges gaps and eliminates irregularities present in the raw point cloud data. The resulting surface offers a more comprehensive and visually intuitive depiction of the 3D scene captured by the point cloud.

Figs. 17–19 demonstrate the surface reconstruction process. Each figure follows a left–right layout. On the left side resides the original point cloud, while the right side showcases the same point cloud after undergoing surface reconstruction using cubic

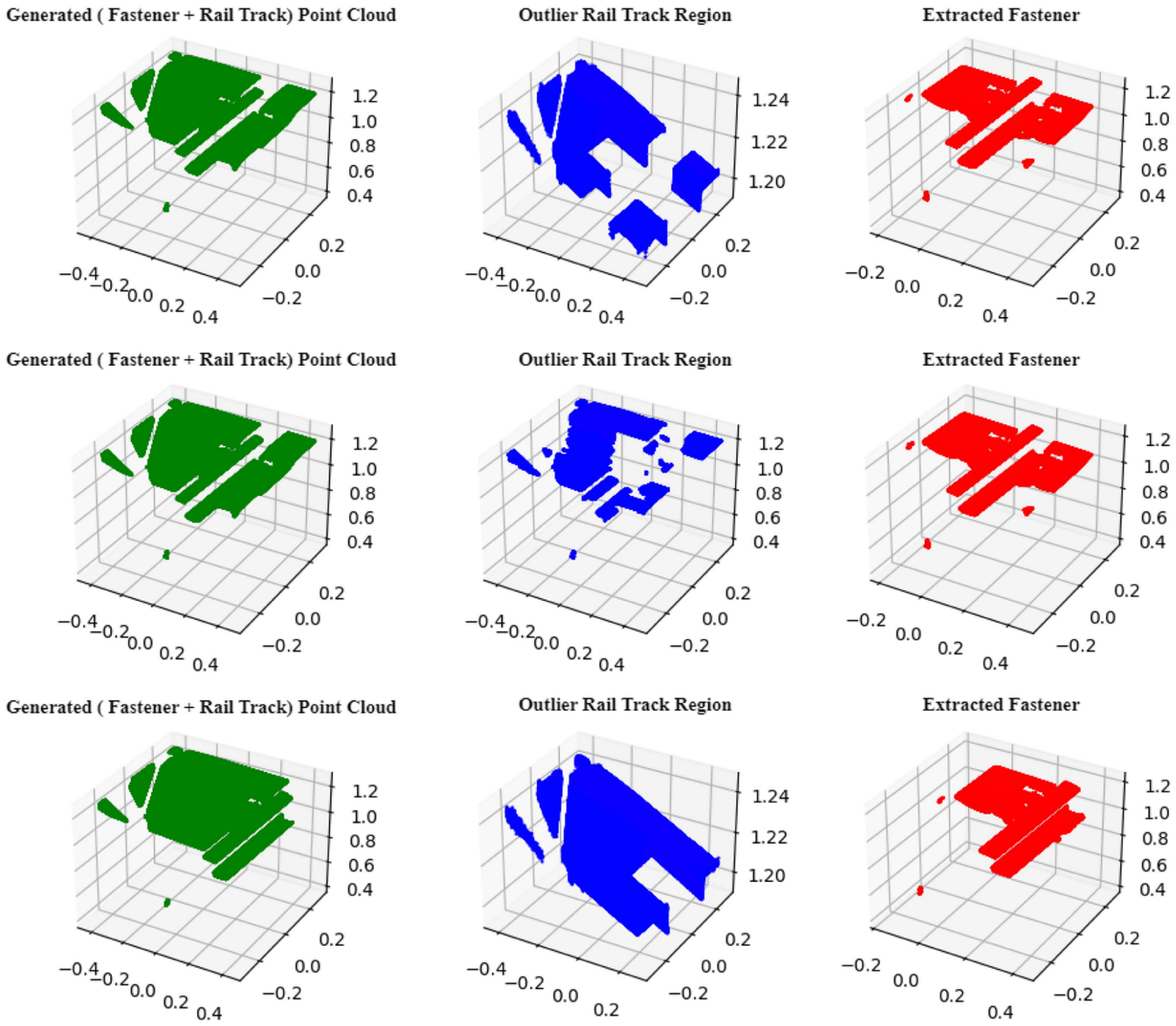


Fig. 16. First columns show the overall generated point clouds, and second columns show the outlier rail track region and third column show the extracted fastener. (The rail track and fastener are separated using the RANSAC algorithm.)

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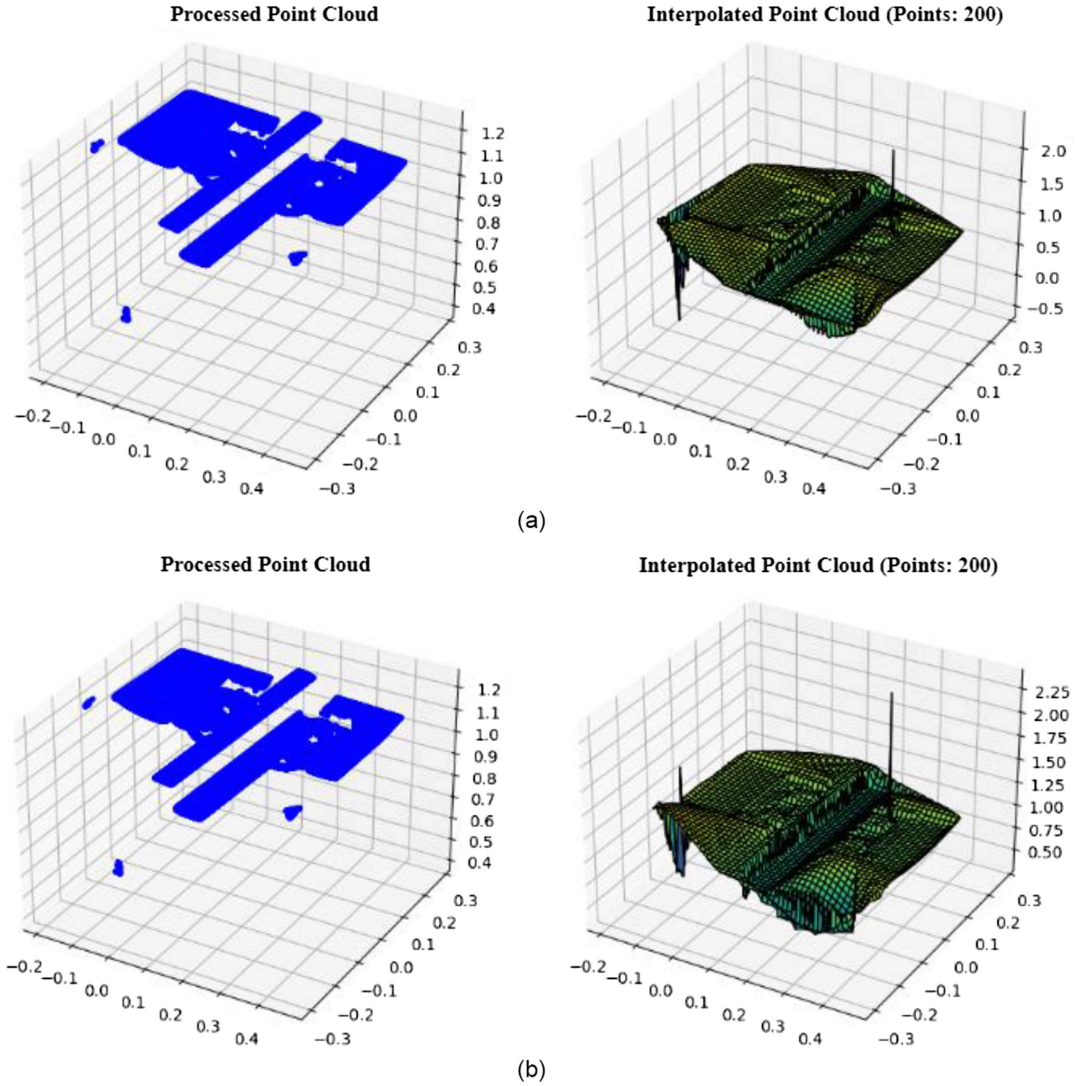


Fig. 17. Left point cloud shows the processed point cloud, while the right point cloud shows 200 points interpolated 3D surface for better visualization: (a) Sample-1; and (b) Sample-2.

interpolation. The captions accompanying each figure specify the number of interpolated points used to generate the surface (e.g., 200 points or 500 points).

A closer examination of the figures reveals the transformative power of interpolation. The raw point cloud often appears fragmented, with gaps and inconsistencies. However, upon applying interpolation, these irregularities disappear. The resulting surface becomes smooth and continuous, offering a more accurate and visually clear representation of the underlying 3D structure as shown in Fig. 20. The benefits of this methodology extend beyond mere visualization. The enhanced point clouds with interpolated surfaces become valuable tools for various applications. In the realm of geographical mapping, these refined point clouds can contribute to the creation of more accurate and detailed maps, facilitating navigation and analysis of terrain features. Similarly, in architectural design, interpolated point clouds can aid in the development of precise 3D models of buildings and structures, enabling architects to visualize and refine their designs with greater clarity.

The impact extends even further into scientific modeling. Researchers across diverse fields can leverage interpolated point clouds to construct intricate 3D models of objects or phenomena under investigation. These models offer unparalleled insights into the spatial relationships and configurations of the elements within the data, fostering a deeper understanding of the scientific principles at play.

Therefore, the combination of segmentation and surface reconstruction empowers us to unlock the hidden details within 3D point clouds. Segmentation allows for the isolation of specific features of interest, while surface reconstruction creates a more informative and visually intuitive representation of the underlying 3D structure. This methodology empowers various disciplines—from geographical mapping and architectural design to scientific modeling—to extract valuable insights and make well-informed decisions based on the captured 3D data. As technology continues to evolve, these techniques will undoubtedly play an increasingly crucial role in shaping our understanding of the world around us.

In the whole sum, firstly we introduce our novel approach starting with self-supervised contrastive learning using a Siamese network. This method efficiently learns representations from image and depth data, facilitating the generation of point clouds. Our primary contribution lies in this representation learning process. Once the point clouds are generated, we employ the RANSAC technique to accurately segment the fastener regions from the entire point cloud, which includes the railway area. This segmentation step

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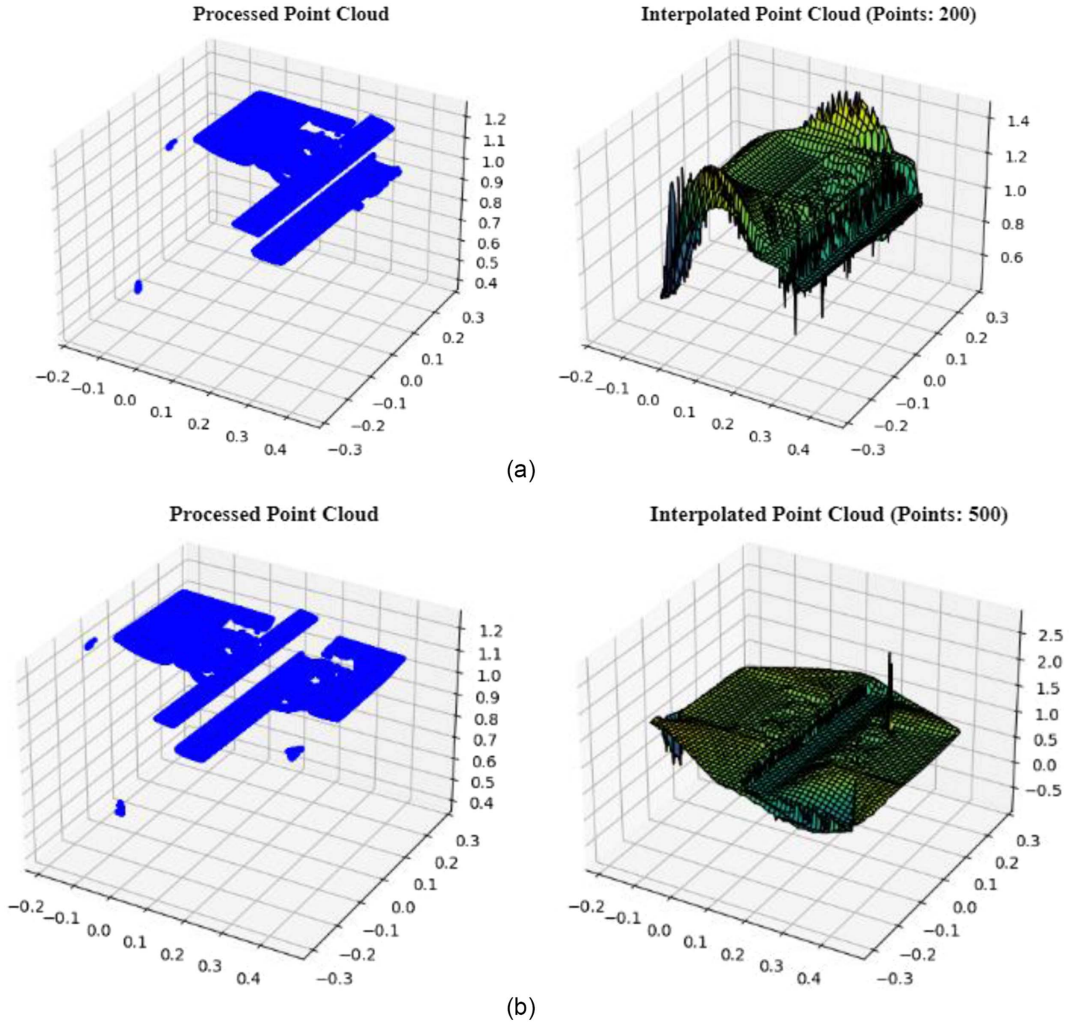


Fig. 18. Left point cloud shows the processed point cloud, while the right point cloud shows points interpolated 3D surface for better visualization: (a) Sample-1 (500 points); and (b) Sample-3 (200 points).

is crucial as it isolates the fasteners for further analysis and inspection. To enhance the visual clarity of the extracted fasteners, we utilize interpolation techniques. This approach ensures that the results are presented with optimal visual fidelity, aiding in the interpretation and validation of our segmentation results. Thus, after self-supervised learning, precise segmentation using RANSAC, and interpolation for visual enhancement are the postprocessing phases that enhanced the overall outcome.

Applications

The enhanced point clouds with interpolated surfaces are valuable tools for visualization, analysis, and decision-making in various fields. These fields include geographical mapping, architectural design, and scientific modeling. The refined representation facilitates a more robust and insightful interpretation of the spatial information within the point cloud data.

Model's Complexity and State-of-the-Art Comparison

This section delves into two key aspects of the proposed lidar simulation framework: its computational complexity and its position within the landscape of existing solutions.

Complexity Analysis

Understanding the computational demands of a model is crucial for practical applications. Here, we meticulously analyze the various modules within our framework to gain insights into their parameter counts and, consequently, their computational burden. The cornerstone of many contemporary models, the transformer module, takes center stage. This module, responsible for processing sequential data, boasts a remarkably low parameter count of 28,976, indicating a lightweight and efficient nature (Fig. 21). This aligns perfectly with the established reputation of transformers for handling sequential data without significant computational overhead. In stark contrast, the multilayer feedforward (MLF) module exhibits a significantly higher parameter count, reaching a total of 850,057 (Table 2). This substantial number, with a vast majority being trainable parameters (849,929), hints at a complex structure likely composed of multiple layers. This complexity allows the MLF module to excel at extracting intricate features and transformations from the data, effectively capturing nuanced patterns that might elude simpler modules.

The Bayesian network module emerges as another focal point, boasting a staggering total parameter count of 4,003,314, all of which are trainable. This high parameter count reflects the intricate probabilistic modeling inherent in Bayesian networks, suggesting a

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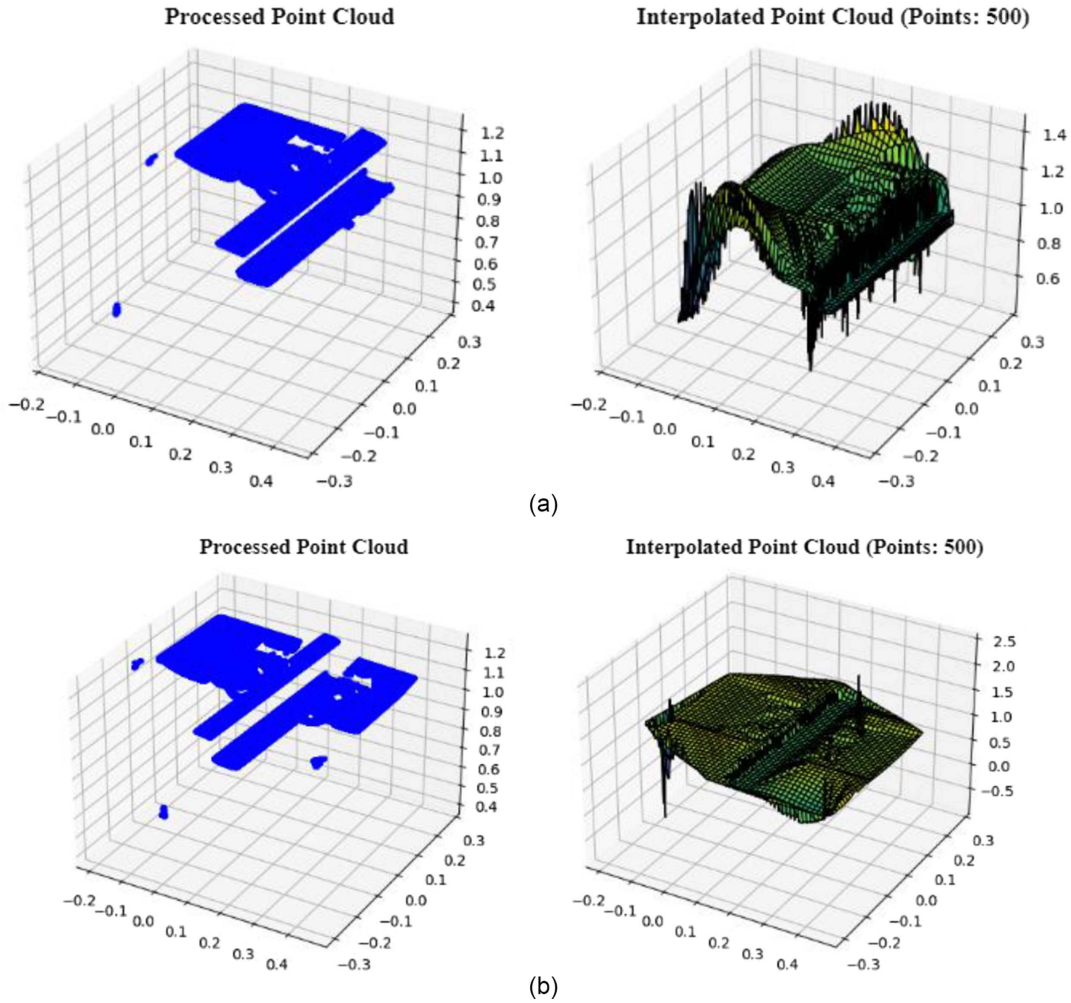


Fig. 19. Left point cloud shows the processed point cloud, while the right point cloud shows 500 points interpolated 3D surface for better visualization: (a) Sample-2; and (b) Sample-3.

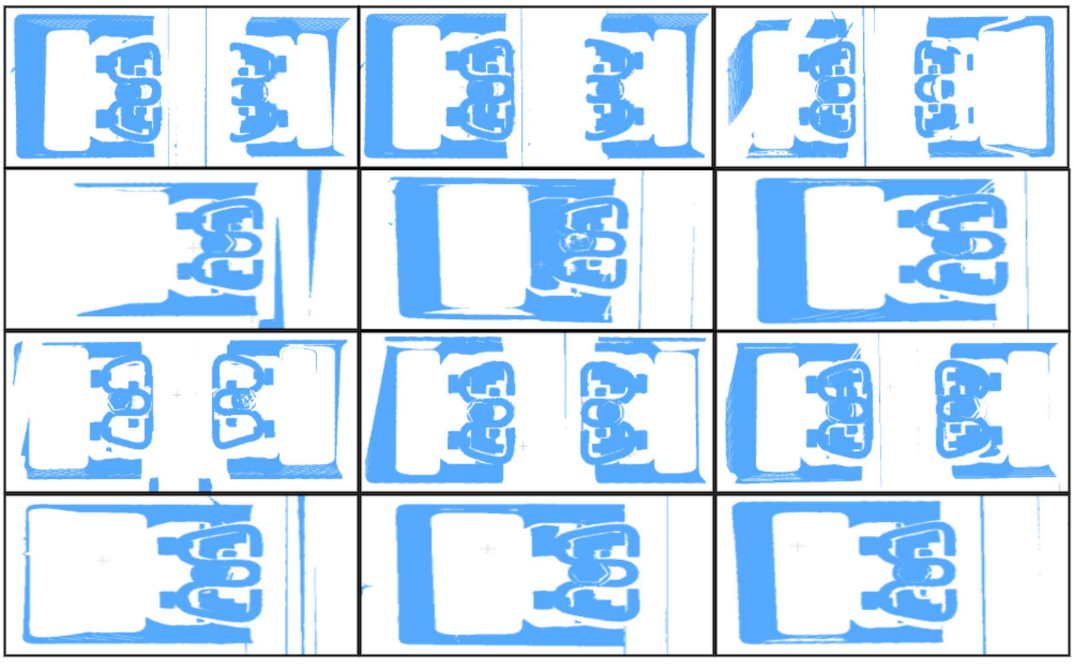


Fig. 20. Segmented fasteners after fasteners point cloud generation.

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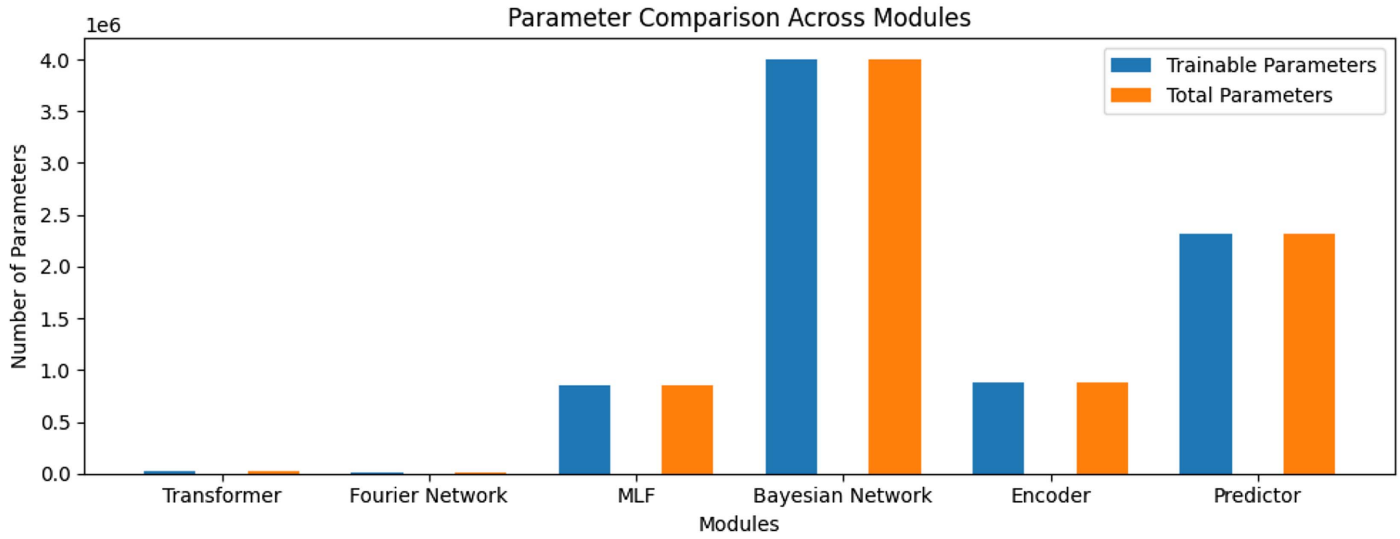


Fig. 21. Parameter comparison.

Table 2. Parameter comparison

No.	Module	Trainable parameters	Nontrainable parameters	Total parameters
1	Transformer	28,976	0	28,976
2	Fourier Network	6,016	0	6,016
3	MLF	849,929	128	850,057
4	Bayesian Network	4,003,314	0	4,003,314
5	Full Encoder	884,921	128	885,049
6	Predictor	2,315,520	0	2,315,520
	Total parameters		**7,203,883 (7.203883 million)**	

sophisticated architecture capable of capturing complex relationships within the data. The sheer volume of parameters underscores its potential for capturing intricate dependencies and uncertainties, making it a key contributor to the overall complexity of the model. Moving on, both the encoder and predictor modules showcase a noteworthy number of parameters, indicating a substantial computational load. The encoder module, a composite structure incorporating the transformer, Fourier network, and MLF modules, has a total parameter count of 885,049, with 884,921 being trainable. The predictor module, on the other hand, is characterized by a total of 2,315,520 parameters, all of which are trainable. It is important to note that the parameters within the encoder module represent a combined contribution from its constituent parts.

The overall model parameter count reaches a significant 7,203,883. However, it is crucial to acknowledge that the model's performance might be impacted by the relatively limited data set size of just 30 samples. To mitigate this potential limitation, the framework employs a self-supervised learning approach. This approach allows the model to extract optimal representations from the available data, even with a smaller data set size. Despite the inherent challenges associated with a limited data set, the self-supervised learning approach is a novel and effective strategy for deriving meaningful insights from such data constraints.

When assessing the model's performance, it is important to recognize the difficulty of achieving exceptionally high results with a limited data set. However, the presented empirical results (in Table 2) demonstrate the model's efficacy across diverse data augmentation techniques. While the limited sample size necessitates a cautious interpretation of performance metrics, the overall approach's novelty and effectiveness are evident in the obtained results. The model's intricate architecture, with each module contributing distinct functionalities, underscores its potential for addressing complex tasks and capturing intricate patterns within the data.

Comparison with State of the Art

This framework stands out as a pioneering solution within the lidar simulation landscape. It is specifically tailored for civil engineering tasks, offering innovative solutions to the challenges inherent in infrastructure monitoring. By leveraging deep-learning techniques such as transformers and Bayesian neural networks for self-supervised contrastive learning, the framework efficiently generates real-time, high-fidelity 3D point clouds from low-dimensional 2D data. This approach fosters automation, accuracy, and efficiency in civil engineering applications. Existing lidar simulators often focus on augmented real data or computer-generated models for applications such as autonomous driving. In contrast, this framework directly addresses the unique requirements of civil engineering tasks, such as rail fastener monitoring. Here, the emphasis lies on accurate geometric reconstructions and real-time insights.

While other lidar simulation approaches, such as those discussed in Fang et al. (2020) and Shan et al. (2020), demonstrate promising performance in their respective domains, they may lack the specificity and tailored solutions required for civil engineering applications. For instance, the study in Fang et al. (2020) proposes a simulator for autonomous driving scenarios, relying on synthetic obstacles augmented into real point clouds. This approach may

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not translate directly to the needs of infrastructure monitoring tasks. Similarly, the framework presented in Zhou et al. (2018) focuses on generating annotated point clouds for 3D obstacle perception in autonomous driving. While prioritizing realism and performance, it might overlook the unique challenges and requirements of civil engineering tasks. Therefore, this article presents a novel and comprehensive solution specifically designed to meet the needs of infrastructure monitoring, contributing significantly to the advancement of the field.

This framework offers several key advantages over existing solutions:

Specificity: Tailored for civil engineering tasks, the framework addresses the unique needs of infrastructure monitoring, such as accurate geometric reconstruction of elements such as rail fasteners and real-time generation of point clouds for immediate analysis.

Efficiency: Leveraging self-supervised learning, the model can effectively extract meaningful insights even from limited data sets, a common challenge in civil engineering applications. This reduces the reliance on vast amounts of labeled data, streamlining the training process.

Real-time Generation: The framework facilitates the generation of high-fidelity 3D point clouds in real-time from low-dimensional 2D data. This enables continuous monitoring and rapid decision-making during infrastructure inspections.

Deep-Learning Techniques: By employing advanced deep-learning techniques such as transformers and Bayesian networks, the framework offers superior capabilities for feature extraction, pattern recognition, and uncertainty modeling within the data. This translates to more accurate and robust simulations.

Overall, this lidar simulation framework presents a significant leap forward in the field of civil engineering. Its ability to address the specific needs of infrastructure monitoring tasks, coupled with its efficiency and real-time capabilities, positions it as a valuable tool for engineers and infrastructure management professionals. As the framework continues to be developed and refined, it has the potential to revolutionize the way infrastructure monitoring is conducted, leading to improved safety, efficiency, and cost-effectiveness in the civil engineering domain.

Step toward Digital Twin

The civil engineering landscape is undergoing a transformative shift with the emergence of real-time 3D point cloud generation, a crucial step toward realizing digital twins (Yan et al. 2023; Zaheer 2024). This research takes center stage in this advancement by introducing a novel framework for the rapid and accurate generation of 3D point clouds specifically for rail fasteners. This framework leverages cutting-edge technologies such as transformers and Bayesian neural networks, akin to a Lidar simulation but with a key distinction: it utilizes a self-supervised contrastive learning approach. This translates to the conversion of low-dimensional data, consisting of images and grayscale depth maps of rail fasteners, into high-fidelity 3D point clouds in real-time. The significance lies not only in the precise geometric reconstructions achieved but also in the framework's ability to learn latent representations, which benefit downstream tasks.

The concept of a digital twin involves creating a virtual representation of a physical system to enhance monitoring, simulation, and optimization processes. In railway engineering, point cloud data are crucial for various applications such as track inspection, maintenance, and infrastructure management. Traditionally, LiDAR systems are employed to collect high-resolution point cloud data of railway tracks and their components. However, the use of LiDAR systems can be both expensive and complex. To address these challenges, we propose a digital twin approach using deep-learning models to generate point cloud data from 2D images and depth information, effectively mimicking the capabilities of a physical LiDAR system.

In the first part of the proposed framework as shown in Fig. 22, illustrated on the left side of the image, a traditional LiDAR system used in the railway domain is depicted. This system comprises several key components: a laser scanner that captures precise 3D measurements of the railway environment, a panoramic camera providing a wide-angle view of the track, a tablet and an inertial measurement unit (IMU) integrated for system control and orientation data, and a track inspection car that serves as a mobile platform equipped with various sensors, including a GNSS receiving antenna for accurate positioning and an odometer for distance measurement. This setup enables comprehensive data collection of railway tracks, capturing detailed point clouds essential for analysis and maintenance. On the right side of the image, the proposed digital twin methodology is illustrated. This approach leverages deep learning to generate point cloud data from 2D images and depth information. The process begins with the preparation of input data, which includes depth images and RGB images. Depth images provide distance information for each pixel, while RGB images offer visual details. Both types of images undergo data augmentation to improve the robustness and generalization of the model. The augmented depth and RGB images are processed through separate multimodal fusion networks (MFNs), which effectively learn to fuse information from both modalities. The fused data from the MFNs is then fed into a Bayesian neural network, which generates the point cloud data. This network incorporates uncertainty estimation, enhancing the reliability and accuracy of the generated point clouds. The model is trained using a composite loss function that includes terms for the outputs of the MFNs, the predictions of the Bayesian neural network, and an additional Bayesian term to handle uncertainty. Moving to the bottom section of the image, we suggest a digital twin methodology for future work, which involves the use of advanced foundation models such as depth estimation models (e.g., MiDaS) to generate depth maps from 2D images. This approach would enable the extraction of depth information without the need for physical LiDAR equipment, significantly reducing costs and complexity. The generated depth maps, combined with RGB images, would then be processed through the described deep-learning pipeline to generate point cloud data. This architecture employs an online network for training and a target network for stabilizing the learning process, with networks being updated progressively to ensure consistent learning. This proposed future work aims to utilize state-of-the-art depth estimation models to further improve the efficiency and accuracy of point cloud data generation. The applications of the generated point clouds in the context of railway engineering are extensive. Specifically, for railway fasteners, the generated point clouds can be used for fastener tightness analysis, ensuring that fasteners are properly secured by analyzing the point cloud data. Additionally, wear and tear detection can be conducted to identify signs of damage on fasteners, enabling timely maintenance. The generated point clouds can also be used for installation verification, checking that fasteners are correctly installed according to predefined standards. Beyond fasteners, point clouds can be employed for alignment and leveling of tracks, rail profile measurement to detect wear and deformation, and erosion monitoring to prevent potential hazards near the tracks. These insights have the potential to enhance safety, reliability, and cost-effectiveness across various civil engineering applications.

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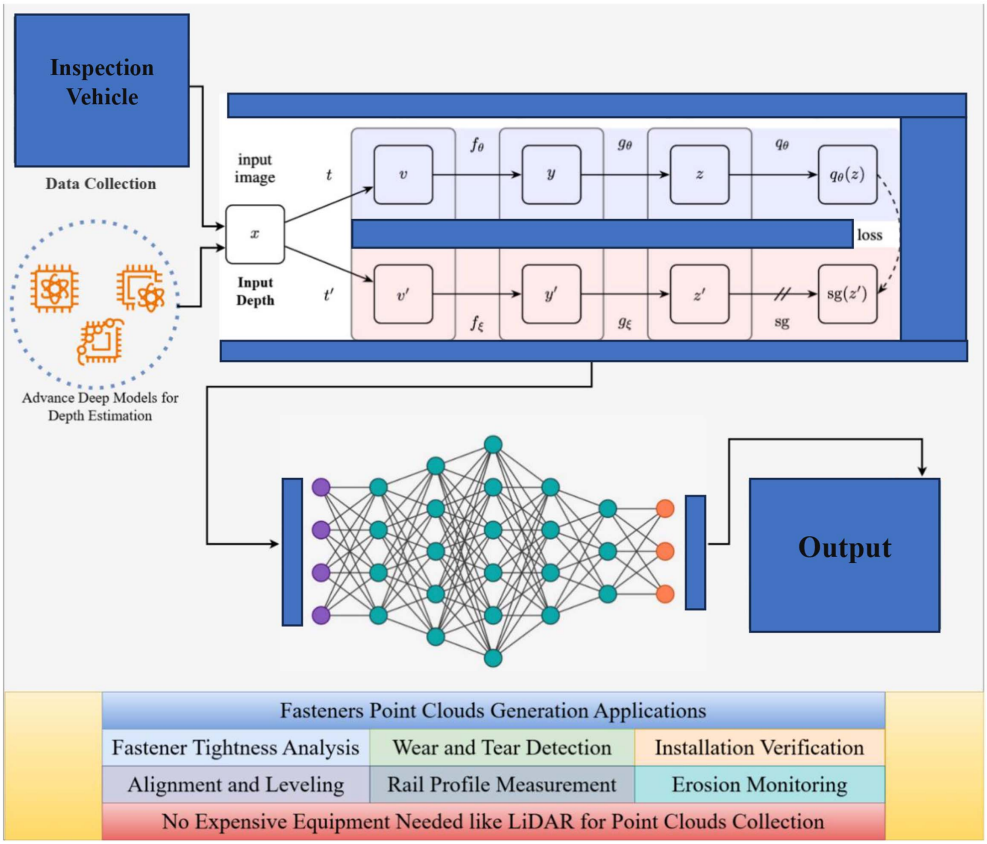


Fig. 22. Theoretical proposed digital twin for fasteners point cloud generation and infrastructure maintenance.

Conclusion

The proposed framework marks a significant step forward in civil engineering and infrastructure monitoring. It overcomes the limitations of traditional LiDAR-based methods by introducing an innovative approach to quickly generate real-time 3D point clouds of rail fasteners. By using advanced deep-learning techniques, specifically transformers and Bayesian neural networks with self-supervised contrastive learning, it effectively converts 2D images and grayscale depth maps into detailed 3D models. One of the key strengths of this framework is its practicality and cost-efficiency. It uses easily accessible 2D images, bypassing the need for costly LiDAR equipment, and addresses common issues such as noise and incomplete data. Its ability to create accurate geometric reconstructions and understand latent data patterns not only enables real-time monitoring but also supports further automated tasks, streamlining civil engineering processes. Moving forward, it is essential to conduct a thorough evaluation of the framework's performance, including comparisons with traditional methods and a detailed analysis of the results. Additionally, incorporating anomaly detection algorithms to measure improvements in the early detection of rail fastener failures will further validate its effectiveness. This framework has the potential to revolutionize infrastructure maintenance by providing real-time insights into the condition of critical components. This paves the way for the development of digital twins, significantly enhancing safety, reliability, and cost-efficiency across various applications. Real-time monitoring supports proactive maintenance strategies, reducing the risk of unexpected failures and accidents. However, there are still challenges to address. The accuracy of the 3D point clouds depends on the quality of the input images and depth maps, and performance may vary with different types of rail fasteners and environmental conditions. Future research should focus on integrating additional data sources and refining the learning algorithms to improve robustness and adaptability. Long-term field tests are also necessary to fully validate the framework's effectiveness in diverse real-world scenarios.

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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